

Domestic Transportation Infrastructure and Gains from Trade in the Vertical Linkages*

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Abstract

Using a two-country Melitz model, we analyze how China's massive railway expansion generates trade gains via vertical linkages. Domestic gains stem from both a composition effect in which improved domestic market access enabled by railway encourages downstream firms to source more inputs domestically and hence raise the domestic value added ratio (DVAR), and a scale effect in which better market access boosts the output level. Intermediate good producers in the foreign country also benefit despite the composition effect, as the scale effect associated with expanded demand from Chinese firms dominates. We test the theoretical predictions with panel data on Chinese manufacturing firms in 2000-2007, addressing endogeneity using a control function approach. Our main empirical findings confirm: (1) Improved upstream access increases exporters' DVAR (explaining 11.57% of its interquantile variation) along with higher value added and revenue; (2) Better access to downstream buyers boosts upstream intermediate producers' value added, profits, and revenue; (3) Foreign intermediate good producers benefit from increased import varieties and quantities of Chinese firms.

JEL classification: F1, R4

Keywords: railway infrastructure, domestic value added ratio, intermediate inputs

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1 Introduction

A prominent feature of China’s development process is its massive investment in transportation infrastructure, with railway construction standing out as one of the core pillars. By 2024, China’s rail network had reached 162,000 km (second only to the United States), while its freight and passenger turnover ranked first globally¹. For a large country like China, railways are uniquely critical for domestic trade: they offer lower transportation costs than roads over long distances, directly boosting the connectivity of the domestic market and the circulation of intermediate goods. Another important, albeit less noted, feature of China’s economy is that its exporters have used more domestic intermediate inputs and become less dependent on imported inputs since China’s accession to the World Trade Organization (WTO) in 2001 (Kee and Tang, 2016). In Global Value Chains (GVC) terminology, the increase in usage of domestic inputs means that the domestic value added ratio (DVAR), the share of domestic value added in gross exports, has increased steadily. This is exceptional: Most countries see a decrease in DVAR during trade liberalization (Johnson and Noguera, 2017), as they tend to import more foreign intermediate goods. A higher DVAR is economically significant, as it signals greater trade gains. The higher the DVAR, the more effectively exports contribute to GDP.

However, the existing literature has not explored a key question: Has China’s massive railway infrastructure buildup contributed to this exceptional rise in DVAR by enhancing access to domestic intermediate goods within vertical trade linkages? Answering this question will not only clarify the link between domestic infrastructure and trade gains but also provide actionable insights for other developing countries interested in emulating China’s infrastructure-driven development path.

To fill this gap, the current study investigates whether expansion of the domestic railway network can increase DVAR among downstream firms in China and generate trade gains for upstream firms in China and abroad, all from the perspective of vertical trade linkages. To understand the underlying mechanisms, we develop a Melitz model in which final good firms (downstream firms) endogenously select intermediate inputs from both domestic and foreign intermediate good firms (upstream firms), subject to both domestic and foreign trade costs.

¹Data sources are the People’s Daily (https://www.gov.cn/lianbo/bumen/202507/content_7033080.htm) and the Ministry of Transportation of China (<https://www.mot.gov.cn/tongjishuju/tielu/202508/P020250827572431120225.pdf>).

Notably, the expansion of the domestic railway network reduces domestic trade costs, but not foreign trade costs.

Our model yields three key predictions. First, DVAR increases due to a composition effect: the domestic railway expansion reduces trade costs, which, in turn, improves market access for domestic intermediate inputs more than for foreign ones. This relative advantage incentivizes downstream firms to source more from domestic suppliers, lifting DVAR. Second, reduced trade costs also generate a scale effect: Lower trade costs increase both the variety of domestic and foreign intermediates used by downstream firms and the quantity of each intermediate good, expanding the overall input base for production. Consequently, vertical trade gains in terms of increased production accrue to downstream and upstream firms, including foreign upstream producers of intermediate goods.² Third, the impact of domestic trade cost reduction on DVAR is spatially concentrated: DVAR rises more sharply for firms in port cities than non-port cities as the relative cost reduction of domestic versus imported inputs is more substantial for port firms; it is also predicted to improve more significantly for firms in highly agglomerated cities under the assumption that access to intermediate goods and agglomeration are complements in strengthening the advantage of home intermediate good firms.

We test the theoretical predictions about trade gains using firm-level and customs transaction data from Chinese firms spanning 2000 to 2007. Using information on the Chinese railway network, we constructed indices for upstream market access (UMA) and downstream market access (DMA) for upstream suppliers and downstream firms, respectively. To address potential endogeneity in market access, we implemented the control function approach proposed by Borusyak and Hull (2023).

Our empirical analysis confirms the theoretical predictions and demonstrates that railway-induced improvements in market access generate trade gains for three distinct sets of firms, with spatial concentration of gains. First, consistent with the predicted composition effect in the model, upstream market access (UMA) has a positive impact on exporters' DVAR. Specifically, the interquartile change in UMA explains approximately 11.57% of the interquartile range of DVAR. Concurrently, the value added and revenue of exporting firms increase with UMA,

²For foreign producers of intermediate goods, the scale effect associated with the expanded demand from Chinese downstream firms dominates the negative impact from the composition effect in which final good firms in China shift expenditure to domestic inputs.

reflecting trade gains from expanded production (a scale effect). Second, better downstream market access (DMA) boosts the value added, profit, and revenue of domestic upstream producers of intermediate goods. This aligns with the theoretical prediction that reduced trade costs generate both a composition effect and a scale effect, which jointly drive output growth for domestic upstream firms. Third, foreign intermediate producers also benefit, as Chinese firms import intermediates in greater varieties and larger quantities. This verifies the model’s prediction that the positive scale effect dominates the negative impact of the composition effect for foreign suppliers. Lastly, trade gains for downstream firms are found to be concentrated, as increases in DVAR and value added are confined to exporter firms in port cities.

The current study contributes to three distinct literatures. First, it relates to a large strand of literature that examines the benefits and costs of vertical trade, with a focus on imported intermediate inputs facilitated by trade liberalization. Observing that U.S. firms tend to offshore non-skill-intensive activities, Feenstra and Hanson (1996) demonstrate that imported intermediate inputs—which the authors and subsequent researchers use to measure outsourcing—significantly increased the demand for skilled labor and increased the share of skilled workers in total wages relative to unskilled labor. The adverse effect of imported inputs on wages and labor market outcome of low-skilled workers are reported by Hummels et al. (2014) with firm-level data from Denmark. By contrast, other studies highlight positive impacts: Amiti and Konings (2007) find that imported inputs improve firm-level productivity in Indonesia, while Goldberg et al. (2010) show that they boost product innovation among domestic firms in India. Beyond firm-level outcomes, Blaum et al. (2018) note that imported inputs deliver substantial benefits to consumers, as they enable firms to reduce production costs and, in turn, lower output prices.

Relative to this strand of literature which focuses on how international trade shapes outcomes through imported intermediate inputs, our paper makes two key extensions by shifting attention to domestic trade costs. First, we emphasize a dual-channel mechanism through which infrastructure reduces domestic trade costs: it improves access to domestic intermediates relative to foreign ones, triggering a composition effect (downstream firms shift expenditure toward domestic inputs) and a scale effect (expanded input variety and quantity boost production). Second, we extend the analysis of vertical trade gains beyond “import-driven changes” to include gains from domestic input market integration: both downstream firms and upstream

firms the dual mechanism, highlighting the broader distribution of gains from domestic trade cost reduction in vertical production linkages.

In terms of the theoretical focus on firm-level input sourcing behavior, our paper is particularly related to Antras et al. (2017), who develop a multi-country model where firms endogenously select the set of economies from which to source intermediate inputs. While Antras et al. (2017) focus on global sourcing decisions across national borders, our study is focused on the implications of domestic trade costs. To this end, we adopt a distinct analytical setup that incorporates not only international trade between two countries but also domestic trade across subnational regions.

Second, we add to the growing literature on the DVAR, a key indicator of trade gains amid the GVC. Since Upward et al. (2013) first proposed the measurement of DVAR, subsequent studies have explored the determinants of DVAR, covering factors such as trade and investment liberalization, tax policies, foreign direct investment (FDI) and the construction of high-speed railways (Kee and Tang, 2016; Lu et al., 2022; Wu et al., 2021; Kee et al., 2024). This paper extends this line of research by focusing on how reductions in domestic trade costs—driven by the expansion of China’s conventional railway network—cause a rise in firms’ DVAR. Although we share the interest of Kee et al. (2024) in China’s railway network, a crucial distinction lies in the type of railway infrastructure analyzed: they focus on high-speed rail, while we examine conventional railways. The former primarily facilitates the transportation of passengers and service flows, while the latter primarily handles most of the rail-based freight transportation and is thus more directly related to the domestic trade of intermediate inputs underlying the dynamics of DVAR.

Finally, our study contributes to the literature on the economic impact of domestic transportation infrastructure. Existing research has documented the broad economic effects of domestic transportation infrastructure, including enhancing export performance (Martincus and Blyde, 2013; Martincus et al., 2017), fostering the agglomeration of economic activities (Henderson et al., 2018), expanding domestic and international trade flows (Donaldson, 2018), strengthening the links between buyer and supplier networks (Bernard et al., 2019), lowering food costs (Porteous, 2019), raising income levels (Donaldson, 2018; Banerjee et al., 2020), and improving farmer productivity and welfare (Sotelo, 2020). Relative to this body of work, a key distinction

of our paper lies in its explicit focus on intermediate goods trade, a channel that generates mutual gains for both upstream and downstream firms.

The subsequent sections of the paper are structured as follows: section 2 presents our theoretical model, while section 3 details our empirical design and data. The main empirical findings and robustness checks are discussed in section 4 and section 5, respectively.

2 Model

To investigate how the expansion of the domestic railway network impacts the utilization of both domestic and foreign intermediate goods in production—and consequently, firms’ DVAR—we develop a two-country Melitz model. In this model, the final good firms (the downstream firms) combine domestic and foreign intermediate goods produced by the downstream suppliers to meet domestic demand.

2.1 Setup

Preference. In the home economy, there exists a continuum of differentiated final goods indexed by j and $j \in [0, 1]$. Similarly, in the foreign economy, there exists a continuum of differentiated final goods indexed by $r \in [0, 1]$.³ Each good j is produced by a single final good firm. To simplify the discussion, we assume that international trade occurs in intermediate goods, but not in final goods.⁴

The representative household in the home country has the following preference:

$$U(\{Q_j\}) = \left[\int_{j=0}^1 Q_j^{(\eta-1)/\eta} dj \right]^{\eta/(\eta-1)}, \quad (1)$$

where η is the elasticity of substitution and Q_j represents the quantity of final good j consumed by the home household. The price of good j is denoted by P_j .

The household budget is equal to the sum of wage income, profits from home firms producing intermediate goods, and profits from home firms producing final goods. The labor supply of the representative household, indicated by L_H , is inelastic.

³Since the focus is on home firms’ choice of intermediate goods, we specify the supply of intermediate goods by foreign firms but suppress other details of the foreign economy, which are symmetric to those of the home economy.

⁴In a model with trade in final goods, the DVAR would remain unchanged as it is determined by productivities of home and foreign intermediate good firms and trade costs.

Based on the preference, the home demand for good j is expressed as follows:

$$\frac{Q_j}{Q} = \left(\frac{P_j}{P} \right)^{-\eta}, \quad (2)$$

where Q and P represent the aggregate indices of the final goods and their prices, respectively.

They are defined as follows:

$$Q = \left[\int_{j=0}^1 Q_j^{(\eta-1)/\eta} dj \right]^{\eta/(\eta-1)}, \quad (3)$$

$$P = \left(\int_{j=0}^1 P_j^{1-\eta} dj \right)^{\frac{1}{1-\eta}}. \quad (4)$$

Production of final goods. The final good firm j uses a set of intermediate goods to produce good j , with the production function defined as follows:

$$Q_j = \left(\int_{i=0}^N X_{ji}^\rho di \right)^{1/\rho}, \quad (5)$$

where X_{ji} denotes the quantity of intermediate good i used by final good firm j , with a price of P_i . N represents the measure of intermediate goods utilized, which is determined endogenously within the model.

The cost minimization for final good firm j implies that the optimal choice of intermediate good i is as follows⁵:

$$X_{ji} = Q_j \frac{P_j}{C_j} \left[\frac{P_i}{C_j} \right]^{\frac{1}{\rho}-1}, \quad (6)$$

where C_j , the cost index for final good j , is given by

$$C_j = \left[\int_{i=0}^N P_i^{\frac{\rho}{\rho-1}} di \right]^{\frac{\rho-1}{\rho}}. \quad (7)$$

The expenditure of final good firm j on intermediate good i and the total expenditure of final good firm j on intermediate goods are

$$e_{ji} = P_i X_{ji} = e_j \left(\frac{P_i}{C_j} \right)^{\frac{\rho}{\rho-1}}, \quad (8)$$

and

$$e_j = P_j Q_j = \int_{i=0}^N e_{ji} di. \quad (9)$$

⁵See Online Appendix A for details of derivations.

Consequently, the total income from the home economy of the intermediate good firm i is as follows:

$$R_i = \left[\int_{j=0}^1 e_{ji} dj \right]. \quad (10)$$

Production of intermediate goods. For any intermediate good i , there is precisely one domestic firm and one foreign firm that can produce it. In both home and foreign countries, the marginal cost of producing intermediate good i is constant. The home and foreign intermediate good firms must pay a fixed cost f_e to enter the market, after which they can learn their respective productivity levels ϕ_H and ϕ_F , which are drawn from the same distribution $\mu(\phi)$. For domestic intermediate good firms, they incur a fixed cost f_{hp} for production in the home economy and a fixed cost f_{fp} for production in the foreign economy. Given the productivity drawn ϕ_{Hi} and the demand of the home and foreign final good firms ($\int_{j=0}^1 X_{ji} dj + \int_{r=0}^1 X_{ri} dr$), the labor hired by intermediate-good firm i (denoted as l_{Hi}) is as follows.⁶

$$l_{Hi} = \frac{\int_{j=0}^1 X_{ji} dj + \int_{r=0}^1 X_{ri} dr}{\phi_{Hi}} + f_{hp} + f_{fp}. \quad (11)$$

Let π_{Hi} and π_{Fi} denote the post-entry profit of intermediate good firm i in the home and foreign markets, respectively. The maximization problems of the profit of the firm i in the home and foreign economies are given respectively by:

$$\begin{aligned} \max_{P_{Hi}} \pi_{Hi} &= \int_{j=0}^1 P_{Hi} X_{ji} dj - \frac{1}{\phi_{Hi}} \int_{j=0}^1 X_{ji} dj \\ &= \int_{j=0}^1 P_{Hi} Q_j \left(\frac{P_{Hi}}{C_j} \right)^{\frac{1}{\rho-1}} dj - \frac{1}{\phi_{Hi}} \int_{j=0}^1 Q_j \left(\frac{P_{Hi}}{C_j} \right)^{\frac{1}{\rho-1}} dj, \end{aligned} \quad (12)$$

$$\begin{aligned} \max_{P_{Hif}} \pi_{Fi} &= \int_{r=0}^1 P_{Hif} X_{ri} dr - \frac{1}{\phi_{Hi}} \int_{r=0}^1 X_{ri} dr \\ &= \int_{r=0}^1 P_{Hif} Q_r \left(\frac{P_{Hif}}{C_r} \right)^{\frac{1}{\rho-1}} dr - \frac{1}{\phi_{Hi}} \int_{r=0}^1 Q_r \left(\frac{P_{Hif}}{C_r} \right)^{\frac{1}{\rho-1}} dr, \end{aligned} \quad (13)$$

where the second equality follows from equation (6); and P_{Hi} and P_{Hif} are the factory gate prices set by domestic producer of intermediate good i for home and foreign final good firms, respectively. The maximization problem yields the usual pricing rule:

$$P_{Hi} = P_{Hif} = \frac{1}{\rho \phi_{Hi}}. \quad (14)$$

⁶There are three scenarios for a intermediate good firm in the home economy: if the marginal cost is high, $X_{ji} = 0$ and $X_{ri} = 0$; if the marginal cost is in a intermediate level, only $X_{ri} = 0$, such that demand is solely domestic; and if the marginal cost is low, both $X_{ji} > 0$ and $X_{ri} > 0$, with demand encompassing both domestic and foreign sources. We assume that no additional fixed costs are incurred for exporting.

Using the last line, the firm's post-entry profit in home and foreign economies can be written as follows.

$$\pi_{Hi} = \int_{j=0}^1 Q_j C_j^{\frac{\rho}{1-\rho}} dj (\rho \phi_{Hi})^{\frac{\rho}{1-\rho}} (1-\rho) - f_{hp}. \quad (15)$$

$$\pi_{Fi} = \int_{r=0}^1 Q_r C_r^{\frac{\rho}{1-\rho}} dr (\rho \phi_{Hi})^{\frac{\rho}{1-\rho}} (1-\rho) - f_{fp}. \quad (16)$$

The measure of intermediate goods in equilibrium is determined by the expected zero-profit condition prior to entry: $E(\pi_{Hi}) + E(\pi_{Fi}) = f_e$.

Final good firms must pay iceberg trade costs for home- and foreign-supplied intermediate goods. To simplify the derivation, we assume a simple geographic structure such that in each country, the production of all intermediate goods occurs in one city, while the production of all final goods occurs in another city. The geographic structure is illustrated in Figure 1. Under this assumption, the trade costs of home-supplied intermediate goods are identical for all varieties, and the same is true for trade costs of foreign-supplied intermediate goods. We denote the pair of costs of trading home and foreign intermediate goods as τ_H and τ_F , respectively.

Let P_{Fi} denote the price at the factory gate of foreign producers of intermediate good i . To minimize the cost of intermediate good i , the final good firms in both countries compare the user costs of intermediate good i supplied by the domestic and foreign producer, $(1 + \tau_H)P_{Hi}$ and $(1 + \tau_F)P_{Fi}$. The user price in the home country for intermediate good i is equal to the minimum of the user costs, that is,

$$P_i = \min\{(1 + \tau_H)P_{Hi}, (1 + \tau_F)P_{Fi}\}. \quad (17)$$

The pattern of production of intermediate goods is determined by the relative production costs and the relative shipping costs. First, if the relative international shipping cost τ_F/τ_H is sufficiently high, then intermediate good producers i in both countries are shielded from international competition. In this case, there is no international trade in intermediate good i , and a country's demand for intermediate goods is served by its domestic firms. Second, if the relative productivity ϕ_{Hi}/ϕ_{Fi} strongly favors the home producer, then the home producer of intermediate good i will also sell to the foreign market, resulting in zero output from the foreign counterpart. Third, if the relative productivity ϕ_{Hi}/ϕ_{Fi} strongly favors the foreign producer, the foreign producer of intermediate good i will capture both markets, leaving the home counterpart with zero output.

We can write the ratio of the user cost of home intermediate goods i to that of the foreign intermediate good i for the home producer of goods j as follows:

$$A_{ji} \equiv \frac{1 + \tau_H}{1 + \tau_F} \times \frac{P_{Hi}}{P_{Fi}}. \quad (18)$$

We denote the trade-cost-adjusted productivity of the home intermediate good firm i as $\hat{\phi}_{Hi} = \frac{\phi_{Hi}}{1 + \tau_H}$, with its cumulative distribution function (c.d.f.) given by $\hat{\phi}_{Hi} \sim \mu_H(\hat{\phi})$. Similarly, the trade-cost-adjusted productivity of the foreign intermediate good firm i is $\hat{\phi}_{Fi} = \frac{\phi_{Fi}}{1 + \tau_F}$, and its c.d.f. is $\hat{\phi}_{Fi} \sim \mu_F(\hat{\phi})$.

The trade-cost-adjusted productivity that prevails in the home market can be written as

$$\hat{\phi}_i = \max\left\{\frac{\phi_{Hi}}{1 + \tau_H}, \frac{\phi_{Fi}}{1 + \tau_F}\right\}. \quad (19)$$

Assuming that the distributions of trade cost adjusted productivity for home and foreign intermediate good firms are independent, the c.d.f. of $\hat{\phi}_i$ is given by:

$$\hat{\phi}_i \sim \nu(\hat{\phi}_i) = \mu_H(\hat{\phi}_i) \cdot \mu_F(\hat{\phi}_i). \quad (20)$$

The corresponding effective user price of intermediate good i in the home country is given by $P_i = \frac{1}{\rho \hat{\phi}_i}$ and its c.d.f. is:

$$P_i \sim 1 - \nu\left(\frac{1}{\rho P_i}\right). \quad (21)$$

Aggregation. Given N , the measure of intermediate goods and the productivity distribution $\nu(\hat{\phi}_i)$, the aggregate price P_j is given by:

$$P_j = C_j = \left(\int_0^\infty P_i^{\frac{\rho}{\rho-1}} N \nu_i(\hat{\phi}_i) d\hat{\phi}_i\right)^{\frac{\rho-1}{\rho}}. \quad (22)$$

Using the pricing rule in equation (14), we obtain:

$$C_j = N^{\frac{\rho-1}{\rho}} P_i(\tilde{\phi}_i) \quad (23)$$

where $\tilde{\phi}_i$ is a weighted average of the firm-level productivity adjusted for trade costs ($\hat{\phi}_i$). The value of $\tilde{\phi}_i$ is independent of N , and its expression is given by:

$$\tilde{\phi}_i = \left[\int_0^\infty \hat{\phi}_i^{\frac{\rho}{1-\rho}} \mu(\hat{\phi}_i) d\hat{\phi}_i\right]^{\frac{1-\rho}{\rho}}. \quad (24)$$

Thus, the total expenditure of final good firm j on all intermediate goods can be written as:

$$e_j = Ne_{ji}(\tilde{\phi}_i). \quad (25)$$

Determination of the set of intermediate goods. It is natural to consider the case in which domestic and foreign intermediate good firms make entry decisions independently. Using equations (15) and (16), for domestic intermediate good firm i , the Zero Cutoff Profit (ZCP) conditions in the home and foreign economies are respectively given by:

$$\pi_{Hi}(\phi_{Hi}) = \int_{j=0}^1 Q_j C_j^{\frac{\rho}{1-\rho}} dj (\rho \hat{\phi}_{Hi})^{\frac{\rho}{1-\rho}} (1-\rho) - f_{hp} = 0, \quad (26)$$

$$\pi_{Fi}(\phi_{Hi}) = \int_{r=0}^1 Q_r C_r^{\frac{\rho}{1-\rho}} dr (\rho \hat{\phi}_{Hi})^{\frac{\rho}{1-\rho}} (1-\rho) - f_{fp} = 0. \quad (27)$$

Let $\hat{\phi}_{Hi}^*$ denote the cutoff productivity that satisfies the ZCP in equation (26) and $\hat{\phi}_{Hif}^*$ denote the cutoff productivity which satisfies the ZCP in equation (27). From equation (24), the average of the firm-level productivity adjusted for trade costs ($\tilde{\phi}_{Hi}$) is fully determined by the cutoff productivity.

From equation (26), the revenue from the home economy of a domestic intermediate-good firm at the cutoff productivity is given by:

$$R_{Hi}(\hat{\phi}_{Hi}^*) = \frac{f_p}{(1-\rho)}. \quad (28)$$

From equations (10) and (15), the home revenue ratio of domestic intermediate good firms with different productivity levels is determined solely by the ratio of their productivities, which is given by:

$$\frac{R_{Hi}(\hat{\phi}_{Hi}^*)}{R_{Hi}(\tilde{\phi}_{Hi})} = \left(\frac{\hat{\phi}_{Hi}^*}{\tilde{\phi}_{Hi}(\hat{\phi}_{Hi}^*)} \right)^{\frac{\rho}{1-\rho}}. \quad (29)$$

Substituting equations (28) and (29) into (15), we express the average profit of a domestic intermediate good firm i selling to final good firm j —which depends on the cutoff productivity—as follows:

$$\int_{j=0}^1 \bar{\pi}_{Hji} dj = R_{Hi}(\tilde{\phi}_{Hi}) (1-\rho) - f_{hp} = f_{hp} \times \left[\left(\frac{\tilde{\phi}_{Hi}(\hat{\phi}_{Hi}^*)}{\hat{\phi}_{Hi}^*} \right)^{\frac{\rho}{1-\rho}} - 1 \right], \quad (30)$$

Given the expression for the average profit in the last line, we have the following Free Entry Condition (FEC):

$$\int_{j=0}^1 \bar{\pi}_{Hji} dj = \frac{\delta f_e}{1 - G(\hat{\phi}_{Hi}^*)}, \quad (31)$$

where $G(\hat{\phi}_{Hi})$ is the c.d.f. of firm i 's trade-cost adjusted productivity,⁷ and δ is the exogenous survival probability.

Following the solution procedure in (Melitz, 2003), we can determine the measure of domestic intermediate good in home economy (denoted as N_H) from ZCP and FEC.

The measure of home-produced intermediate goods used by home final good firms, denoted as N_H , is

$$N_H = \mathbb{P}(\hat{\phi}_{Hi} > \hat{\phi}_{Fi}) \times \frac{\int_{j=0}^1 e_j dj}{\int_{j=0}^1 \bar{e}_{Hji} dj} = \frac{\mathbb{P}(\hat{\phi}_{Hi} > \hat{\phi}_{Fi}) \times L_H}{\sigma(\int_{j=0}^1 \bar{\pi}_{Hji} dj + f_{hp})}, \quad (32)$$

where $\mathbb{P}(\hat{\phi}_{Hi} > \hat{\phi}_{Fi})$ represents the probability that the productivity of domestic firms is higher than that of foreign firms, i.e., the probability that domestic firms are selected by the home final good firms.⁸

Similarly, the measure of foreign intermediate goods imported into the home economy, denoted as N_F , is

$$N_F = \mathbb{P}(\hat{\phi}_{Fi} > \hat{\phi}_{Hi}) \times \frac{\int_{j=0}^1 e_j dj}{\int_{j=0}^1 \bar{e}_{Fji} dj} = \frac{\mathbb{P}(\hat{\phi}_{Fi} > \hat{\phi}_{Hi}) \times L_H}{\sigma(\int_{j=0}^1 \bar{\pi}_{Fji} dj + f_{fp})}, \quad (33)$$

where $\bar{e}_{Fji}$ and $\bar{\pi}_{Fji}$ represent the average revenue and profit of a foreign intermediate good firm in home economy, respectively.

The total measure of intermediate goods, denoted N , is equal to the sum of N_H and N_F .

$$N = N_H + N_F. \quad (34)$$

To obtain the relationship between N and the average profits of domestic and foreign firms, we substitute equation (32) and (33) into (34). Simplification yields the solution for N in terms of fundamental variables.

$$N = N_H + N_F = \frac{1}{\sigma} \left(\frac{\mathbb{P}(\hat{\phi}_{Hi} > \hat{\phi}_{Fi}) L_H}{(\int_{j=0}^1 \bar{\pi}_{Hji} dj + f_{hp})} + \frac{\mathbb{P}(\hat{\phi}_{Fi} > \hat{\phi}_{Hi}) L_H}{(\int_{j=0}^1 \bar{\pi}_{Fji} dj + f_{fp})} \right), \quad (35)$$

where L_H , $\bar{\pi}_{Fji}$ and $\bar{\pi}_{Hji}$ represent domestic labor, and the average profits of foreign and domestic firms, respectively.

⁷For the sake of simplicity in calculation, we assume that the joint distribution of productivity and trade costs follows a Pareto distribution.

⁸ $\mathbb{P}(\hat{\phi}_{Hi} > \hat{\phi}_{Fi})$ and $\mathbb{P}(\hat{\phi}_{Fi} > \hat{\phi}_{Hi})$ are constants and are related to the lower bound value of the Pareto distribution function as well as its parameters. For specific details, refer to Online Appendix D.

Domestic Value-Added Ratio. The share of imported intermediate goods in the total cost of intermediate goods is expressed as

$$\frac{\int_{i=0}^{N_F} (1 + \tau_F) P_{Fi} X_{Fji} di}{\int_{i=0}^N P_i X_{ji} di}.$$

Substituting equations (6) and (18) into the last line, we can obtain the share of imported intermediate goods in total cost of intermediate goods

$$\frac{\int_{i=0}^{N_F} (1 + \tau_F) P_{Fi} X_{Fji} di}{\int_{i=0}^N P_i X_{ji} di} = \frac{1}{\left(1 + \left(\frac{N}{N_F} - 1\right) \frac{1}{A_{ji}}\right)}. \quad (36)$$

Thus, domestic value-added ratio (DVAR) is given by

$$\begin{aligned} DVAR_j &= 1 - \frac{\int_{i=0}^{N_F} (1 + \tau_F) P_{Fi} X_{Fji} di}{\int_{i=0}^N P_i X_{ji} di} \\ &= 1 - \frac{1}{\left(1 + \left(\frac{N}{N_F} - 1\right) \frac{1}{A_{ji}}\right)} \end{aligned} \quad (37)$$

Reflecting the weight of domestic intermediate inputs in total cost of intermediate inputs, DVAR is an important measure of the composition of intermediate inputs. In addition, because the DVAR measures the fraction of a final good firm's revenue that is recorded in GDP, it captures trade gains per unit of final output that come from both intermediate good production and final good production in the home economy.

Equilibrium. The exogenous primitives of the model are the parameters $(\eta, \rho, \sigma, f_{hp}, f_{fp}, f_e)$, the productivity distributions for home and foreign firms $(\mu(\cdot))$, and the labor supplies in home and foreign countries (L_H, L_F) . Given these primitives, an equilibrium consists of the prices for differentiated intermediate goods $\{P_i\}$, the prices of final goods $\{P_j\}$ and the quantity of final good j consumed by the household $\{Q_j\}$. These prices and allocations must solve the utility maximization of households and the profit maximization problems of firms. Other endogenous variables, such as the cost of final goods j , operating profits per intermediate firm, the expenditure of final good firm j on intermediate good i , the total expenditure of final good firm j , and the measure of home and foreign intermediate good firms, can also be expressed as functions of $\{P_i\}$, $\{P_j\}$, $\{Q_j\}$. Please see Online Appendix B for details about how to solve for the equilibrium.

2.2 Testable Predictions

Using this model, we can obtain predictions about the relationship between domestic trade costs and the gains from trade, as well as the gains for final good firms (downstream firms) and intermediate good firms (upstream firms). We begin with a proposition about DVAR which measure the composition effect of a reduction domestic trade cost.

Proposition 1. *When domestic trade costs decrease, the DVAR of final good firms will increase.*

Proof. Combing equations (33) and (35), we can obtain the expression for the weight of foreign intermediate goods in the total spending of firm j on intermediate goods

$$\frac{N_F}{N} = \frac{\mathbb{P}(\hat{\phi}_{Fi} > \hat{\phi}_{Hi}) \times \left(\int_{j=0}^1 \bar{\pi}_{Hji} dj + f_{hp} \right)}{\mathbb{P}(\hat{\phi}_{Hi} > \hat{\phi}_{Fi}) \times \left(\int_{j=0}^1 \bar{\pi}_{Fji} dj + f_{fp} \right) + \mathbb{P}(\hat{\phi}_{Fi} > \hat{\phi}_{Hi}) \times \left(\int_{j=0}^1 \bar{\pi}_{Hji} dj + f_{hp} \right)}. \quad (38)$$

According to equations (17) and (22), when the domestic trade cost of intermediate goods decreases, the cost index C_j decreases. The Free Entry Condition implies that more domestic intermediate good firms will enter the market when perceiving profitable opportunities. Consequently, according to equations (30) and (31), in the new equilibrium, the average post-entry profit per firm, denoted as $\bar{\pi}_{Hji}$, decreases due to intensified market competition.⁹

Hence, equation (38) implies that when $\bar{\pi}_{Hji}$ decreases, the weight of foreign intermediate goods (N_F/N) decreases. According to equation (18), a reduction in domestic trade costs leads to a decrease in A_{ji} , where A_{ji} represents the ratio of the user cost of domestic intermediate goods i to that of foreign intermediate goods i for producers of final goods j . Therefore, based on equation (37), a decrease in N_F/N and A_{ji} leads to an increase in the DVAR of the final good firms. $\square$

This result aligns with intuition: Reduction in domestic trade costs improves the cost competitiveness of domestic intermediate goods relative to foreign intermediate goods. As a result, final good firms are more inclined to increase the proportion of domestic intermediate goods used in their production processes. The next two propositions outline the adjustments in sourcing behavior that underlie the change in DVAR.

⁹Please see Online Appendix C for details about how trade cost affects post-entry profit in equilibrium.

Proposition 2. *A decrease in domestic trade costs causes final good firms in the home economy to increase the varieties of intermediate goods used in their production and to increase the varieties of imported intermediate goods.*

Proof. As argued in the proof of Proposition 1, when the domestic trade cost of intermediate goods decreases, more intermediate good firms will enter the market in response to perceived profit opportunities. Consequently, according to equation (23), final good firms now use more varieties of intermediate goods, including more varieties of imported intermediate goods, to minimize their costs. Therefore, a decrease in domestic trade costs results in an increase in both N and N_F . $\square$

Proposition 3. *After a decrease in domestic trade costs, final good firms increase the purchase quantity of each intermediate good produced by both domestic and foreign intermediate good firms.*

Proof. According to equation (39), a decrease in domestic trade costs leads to an increase in the optimal output of final good firm j . Consequently, according to equation (6), the quantity of each intermediate good used by final good firm j increases, and the quantity of each intermediate good imported by final good firm j also increases. $\square$

This finding demonstrates that a decrease in the cost of domestic trade also promotes international trade by increasing the use of imported intermediate goods along the intensive margin. Thus, following a reduction of trade cost in the home country, the positive scale effect on foreign intermediate good firms dominates the negative composition effect. The next proposition shows that, in addition to causing changes in sourcing behavior and hence increasing DVAR, a reduction in trade cost also increases revenue of final good firms in the home economy. That is, there is a positive scale effect.

Proposition 4. *A reduction in domestic trade costs leads to an increase in the revenue of final good firms.*

Proof. Recall that e_j represents the total expenditure of final good firm j on intermediate goods. Substituting equations (5), (8), (18), and (25) into $e_j = P_j Q_j$, rearrangement yields

$$Q_j = P_j^{-1} (1 + \tau_H) X_{Hji} A_{ji}^p \left(\int_{i=0}^N P_{Hi} di \right). \quad (39)$$

Thus, the revenue of firm j is given by:

$$R_j = PQ^{1/\eta} Q_j^{(\eta-1)/\eta} = PQ^{1/\eta} (P_j^{-1} (1 + \tau_H) X_{Hji} A_{ji}^\rho (\int_{i=0}^N P_{Hi} di))^{(\eta-1)/\eta}. \quad (40)$$

As shown in the proof of Proposition 1, when domestic trade costs decrease, the average post-entry profit per firm, denoted as $\bar{\pi}_{Hji}$, decreases. According to equation (35), N increases, which ultimately leads to an increase in R_j by virtue of equation (40).

Beyond this channel, the reduction in domestic trade costs further enables final good firms to choose a greater variety of intermediate goods; this, in turn, expands their output level and consequently boosts the revenue. $\square$

The concentration of economic activities is a well-documented fact in spatial and regional economics. In the following two propositions, we examine how market access interacts with two key drivers of concentration: coastal location advantages and agglomeration economies.

To obtain theoretical predictions about how domestic trade costs affect the DVAR of final good firms in port and non-port cities, we modified the spatial structure of the model to include two sites of final good production, a port city and a non-port city (see Figure 2), in each country. Under the structure, the non-port city can only access foreign intermediate good through the port city, thus the trade cost satisfies

$$\tau_F^{NP} = \tau_F^P + \tau_D, \quad (41)$$

where τ_F^{NP} denotes the trade cost for purchasing foreign intermediate goods in the non-port city, τ_F^P the trade cost for purchasing foreign intermediate goods in the port city, and τ_D the trade cost between the port and non-port city. The trade costs for purchasing home intermediate goods in the port and non-port city are τ_H^P and τ_H^{NP} , respectively. Under this spatical structure, we have the following proposition.

Proposition 5. *Following a decrease in domestic trade costs, if reduction in τ_F^{NP} does not dominate the combined reduction in τ_F^P and τ_D , final good firms in the port city will experience a larger increase in DVAR compared to final good firms in the non-port city.*

Proof. Let A_{ji}^P and A_{ji}^{NP} represent the relative user cost of domestic intermediate goods i to the user cost of foreign intermediate goods i for final good firms in port cities and non-port cities,

respectively. In our framework, these pair of relative costs can be written as follows:

$$A_{ji}^P = \frac{1 + \tau_H^P}{1 + \tau_F^P} \times \frac{P_{Hi}}{P_{Fi}}, \quad (42)$$

$$A_{ji}^{NP} = \frac{1 + \tau_H^{NP}}{1 + \tau_F^{NP}} \times \frac{P_{Hi}}{P_{Fi}}. \quad (43)$$

Dividing the second last equation by the last, we obtain

$$\begin{aligned} \frac{A_{ji}^P}{A_{ji}^{NP}} &= \frac{(1 + \tau_H^P)(1 + \tau_F^{NP})}{(1 + \tau_F^P)(1 + \tau_H^{NP})} \\ &= \frac{(1 + \tau_H^P)(1 + \tau_F^P + \tau_D)}{(1 + \tau_F^P)(1 + \tau_H^{NP})}. \end{aligned}$$

Taking the log of the last line yields

$$\begin{aligned} \ln\left(\frac{A_{ji}^P}{A_{ji}^{NP}}\right) &= \ln(1 + \tau_H^P) + \ln(1 + \tau_F^P + \tau_D) - \ln(1 + \tau_F^P) - \ln(1 + \tau_H^{NP}) \\ &\approx \tau_H^P + \tau_F^P + \tau_D - \tau_F^P - \tau_H^{NP} \\ &= \tau_H^P + \tau_D - \tau_H^{NP}, \end{aligned}$$

where the second last line holds if the trade costs are small. It is natural to assume the trade costs are continuous and differentiable functions of railway. Using the notation $[x]'$ to denote a variable's derivative with respect to expansion of railway, we have

$$[\ln(\frac{A_{ji}^P}{A_{ji}^{NP}})]' \approx [\tau_H^P]' + [\tau_D]' - [\tau_H^{NP}]'.$$

As trade costs either decrease or remain constant after a railway expansion, the three derivatives in the right hand side of the last line are all non-positive. It follows that if magnitude of $[\tau_F^{NP}]'$ does not dominate the magnitude of $[\tau_F^P]'$ and $[\tau_D]'$ combined,

$$(\frac{A_{ji}^P}{A_{ji}^{NP}})' > 0.$$

That is A_{ji} decreases by a larger magnitude in the port city than in the non-port city. Applying the argument in Proposition 1, final good firms in the port city will experience a larger increase in DVAR compared to final good firms in the non-port city. $\square$

Note that Proposition 5 holds if railway construction reduces the port and non-port city's trade costs for purchasing home intermediate goods by a similar magnitude, which is arguably plausible. In this case, when domestic trade costs decrease, the total trade cost of foreign

intermediate goods decrease by a greater extent for firms in non-port cities than for those in port cities. Since firms in both types of cities undergo a similar reduction in the shipping cost of domestic inputs, the relative price of domestic to foreign inputs will fall more for firms in port cities than for those in non-port cities. Consequently, improvements in the domestic railway network will provide firms in port cities with a stronger incentive to substitute foreign intermediate goods for domestic ones.

To analyze the joint effect of agglomeration and domestic trade costs, we make the following assumption.

Assumption 1. In cities with a higher degree of agglomeration (denoted g_H), the effect of a decrease in domestic trade costs (τ_H) on the relative user cost of domestic intermediate goods to foreign intermediate goods (A_{ji}) is stronger. Mathematically, the assumption is $\frac{\partial^2 A_{ji}}{\partial g_H \partial \tau_H} > 0$.

The economic essence of this assumption is that access to intermediate goods and agglomeration are complements. A concrete scenario that satisfies this assumption is as follows: while a decrease in domestic trade costs reduces the relative user cost of domestic intermediate goods for all firms, spatially clustered firms will enjoy additional cost savings, as they can easily share information on suppliers among these local peers. We make the assumption to obtain insight and theoretical predictions regarding how agglomeration interacts with domestic trade costs, but our empirical focus is on estimating the joint effect of agglomeration and domestic trade costs on DVAR.

Proposition 6. *If Assumption 1 holds, following a decrease in domestic trade costs, the DVAR of final good firms in cities with a higher degree of agglomeration will have a greater increase.*

Proof. Under Assumption 1, as domestic trade costs decrease, domestic intermediate goods will have greater cost advantages in cities with a higher degree of agglomeration. Namely, A_{ji} decreases by a larger magnitude in these cities. Applying the same argument as in Proposition 1, it follows that the DVAR of the final good firms in cities with a higher degree of agglomeration will experience a greater increase. $\square$

In the next section, we test these propositions with panel data on Chinese manufacturing firms in 2000-2007.

3 Empirical Strategy and Data

3.1 Regression Model and Identification Strategy

Ordinary least squares (OLS) estimates of the effects of market access on DVAR and other outcome variables are highly likely to be biased because market access associated with railway expansion can be endogenous. For example, policymakers may choose to build railways to serve cities with strong economic performance. In such a scenario, holding other factors constant, firms in economically competitive cities are likely to use more domestic intermediate inputs, thus reporting a higher DVAR. The direction of bias can also go the other way if governments choose to build railways to connect remote and less prosperous cities.

Although instrumental variables, especially geographic or predetermined characteristics, are commonly used in the literature to isolate exogenous variation in railway connectivity for 2SLS estimation, Borusyak and Hull (2023) highlight a critical limitation: such approaches may still produce biased estimates of MA's effects. This is because cities with predetermined geographic advantages (e.g., central locations) tend to have better railway access, leading to higher MA levels. To mitigate this bias, we adopt the control function approach developed by Borusyak and Hull (2023). To be specific, the 2SLS framework is modified in two steps: (1) use an instrumental variable to capture exogenous variation in railway construction (and thus MA); (2) augment the second stage regression with the interactions between year dummies and the expected level of MA (EMA_{jc}) in the initial period, defined as the MA predicted for industry j in city c in the initial year based on its geographic characteristics and economic development. This two-step control function approach isolates the exogenous component of MA by leveraging deviations of actual railway connectivity from its expected level. Specifically, interactions between year dummies and EMA_{jc} absorb the nonrandom component of MA driven by persistent geographic or historical factors, acting as a control for confounding geographic advantages. This ensures that the estimated coefficient on MA reflects its true causal effect. By isolating MA's exogenous component and absorbing its geography-driven nonrandom variation, this two-step control function approach ensures that the estimated coefficient on MA reflects its true causal effect.

Thus, our regression model is as follows.

$$Y_{ijct} = \beta_0 + \beta_1 \times \ln(MA_{jct}) + \beta_2 \times \ln(EMA_{jc}) \times \eta_t \\ + \beta_3 \times I_{ijct} + \beta_4 \times X_{ijc} \times \eta_t + \beta_5 \times Z_c \times \eta_t + \varphi_i + \omega_t + \mu_{ijct}, \quad (44)$$

where i , j , c and t are the indices for firm, industry, city, and year. Y_{ijct} is the outcome variables of interest. Our main dependent variable is $DVAR_{ijct}$ as it provides an overall measure of the gains from trade from both intermediate good production and final good production. In addition to DVAR, we also use firm-level value added, profit, and revenue from the main business as outcome variables in regressions.

MA_{jct} is market access index for industry j in city c during year t . Adopting the approach of Donaldson and Hornbeck (2016), we construct separate MA measures from the perspective of downstream and upstream firms: upstream market access (UMA) and downstream market access (DMA). The UMA index measures the degree of market access to upstream inputs from the point of view of downstream firms (final good firms), while DMA captures the access to downstream markets from the point of view of upstream firms (intermediate good firms). The specific construction procedures for DVAR, UMA, and DMA are detailed in subsection 3.2 and subsection 3.3. The variable EMA_{jc} is the expected market access of industry j in city c in the initial period, calculated using the method described in Borusyak and Hull (2023). For upstream and downstream firms, we construct the corresponding expected market access indices which are denoted as $EUMA_{jc}$ and $EDMA_{jc}$.

The vector I_{ijct} includes an indicator for firms engaged in processing trade and an indicator for state-owned enterprises (SOE). The vector X_{ijc} includes the Herfindahl–Hirschman Index of the four-digit China Industry Classification (CIC) industry to which a firm belongs in the initial period and the firm age measured as the logarithm of firm age in the initial period plus 1.

To eliminate the endogenous bias associated with time-varying characteristics at the firm and city levels, we follow the methodology of Lu et al. (2017) to control for the effects of such factors by introducing interaction terms between the initial characteristic variables of firms and cities, X_{ijc} and Z_c , and a year dummy η_t in the regressions. To be specific, the vector Z_c contains initial period values of the logarithm of the GDP of the city, the logarithm of the average wage of the city, the ratio of fixed investment to GDP in the city, the ratio of the total

mileage of the road to the land area in the city, and the logarithm of the rail transport distance from the city to the sea port ¹⁰.

Lastly, ψ_i , ω_t , and μ_{ijct} represent firm fixed effects, year fixed effects, and a random disturbance term, respectively.

3.2 Construction of Firms' Ratio of Domestic Value Added in Exports to Gross Exports (DVAR)

To construct the DVAR variable, we merge firm-level data from the Annual Survey of Industrial Firms (ASIF) with firm-level trade data from China Customs Trade Statistics during 2000-2007, by following the two-step procedure of Upward et al. (2013). First, we use the firm name and year as matching variables in the first stage of matching. Second, we use the last seven digits of the business phone number and the postal code to match additional observations that were not matched in the first step. Lastly, we use the methods of Feenstra et al. (2014) and Yu (2015) to eliminate abnormal observations and observations that lack key information.

In computing the DVAR variable, we begin with the formula of Upward et al. (2013) which assumes that all imported inputs for a firm are intermediate inputs. Specifically, for firms engaged in processing trade, all imported inputs are used in export processing, while for firms involved in ordinary trade, the imported inputs are proportionally allocated between domestic sales and exports. Accordingly, the formula for DVAR is given by:

$$DVAR_u = 1 - \frac{V_{AF}}{X} = 1 - \frac{M^p + X^o[M^o/(D + X^o)]}{X}, \quad (45)$$

where V_F is the foreign value added in the export of a firm, and the variables M , X and D denote the import, export and domestic sale of a firm. The superscripts p and o indicate the processing trade and the ordinary trade, respectively. In the actual calculation, we use the import data obtained from the Customs Office. Based on the Annual Survey of Industrial Firms (ASIF), the domestic sale of a firm is calculated as the difference between the firm's total sales and its exports. For firms reporting sales values smaller than export values, we assume that the foreign value added (V_F) is equal to the sum of imports in processing trade (M^p) and imports in ordinary trade (M^o). In cases where the foreign value added exceeds the export value, we set the firm's DVAR to 0, with the ratio of foreign value added to export set to 1.

¹⁰The railway transportation distance from the city to seaport is measured by the weighted railway transportation distance from the city to the top three nearest ports in the initial period, with weights based on port throughput.

To improve the accuracy of the above method, we follow Kee and Tang (2016) to make three adjustments to the computation process. First, we distinguish between types of import in ordinary trade. By converting the six-digit Harmonized Commodity Description and Coding System (HS) codes into product codes under the Broad Economic Categories (BEC), we are able to identify the ordinary trade import of intermediate goods defined by the BEC (denoted as M_m^o) while excluding imports of consumption goods and capital goods. We then use the import of intermediate goods (M_m^o) to replace the gross import in ordinary trade (M^o) in equation (45). Second, following Ahn et al. (2011), we identify intermediary firms by searching for phrases that indicate their intermediary status. These phrases include “import and export”, “economic and trade”, “trading”, “science and trade”, and “foreign trade”,¹¹ which allows us to further correct the foreign value added (V_F), processing trade imports (M^p), and ordinary trade imports (M^o) in equation (45) to obtain the adjusted foreign value added (denoted as V_{AF}), adjusted processing trade imports (M_A^p) and adjusted ordinary trade imports of intermediate inputs (denoted as M_{Am}^o). Third, we exclude overseas components contained in domestically purchased inputs (γ_{jF}). To quantify $\frac{\gamma_{jF}}{X}$, we use the results of Kee and Tang (2016) whose estimated value of $\frac{\gamma_{jF}}{X}$ ranges from 0.4% to 5.7%, depending on the industry. After the three adjustments, the formula for DVAR becomes

$$DVAR = 1 - \frac{V_{AF}}{X} = 1 - \frac{\{M_A^p + X^o[M_{Am}^o/(D + X^o)]\}}{X} - \frac{\gamma_{jF}}{X}. \quad (46)$$

In the calculation of the DVAR, we exclude two types of firms that are not suitable for inclusion in our study. The first type is excessive importers. These are export-processing firms that import inputs beyond their production needs and sell the excess inputs to other firms. Specifically, we identify excessive importers as export-processing firms with an import-to-export ratio (based on customs office data) greater than the ratio of intermediate inputs to sales (based on ASIF data). We remove these firms because they tend to report negative values of DVAR. The second type is excessive exporters. Excessive exporters purchase inputs imported by the excessive importers, which causes their DVAR, as computed from equation (46), to be inflated toward 1. We identify excessive exporters as export processing firms that report an import-to-export ratio smaller than the 25th percentile of ordinary trade firms in the same industry.

¹¹The corresponding terms in Chinese are “jinchukou”, “jingmao”, “maoyi”, “kemaoyi”, and “waijing”.

3.3 Construction of Market Access Indices and Instrumental Variables

Based on *China Railway Yearbook*, *Chinese Research Data Services Platform* and *China Urban Statistical Yearbook*, we construct two market access indices—upstream market access (UMA) and downstream market access (DMA)—based on connectivity to the railway for 276 prefecture-level cities in China from 2000 to 2007 by applying the method of Donaldson and Hornbeck (2016). We focus on railway-related MA because, according to data from the National Bureau of Statistics of China, railways accounted for approximately 50% of national freight turnover (in ton-kilometers) on average in the 2000s' decade. Furthermore, since railways are the most efficient means of long-distance inland transportation, connectivity to the railways plays a crucial role in determining trade costs between prefectures in a vast country like China.

We modify the formula for market access in Donaldson and Hornbeck (2016) to construct the UMA and DMA indices as follows:

$$UMA_{jct} = \sum_{c' \neq c} \tau_{cc',t}^{-\theta} \sum_{j' \neq j} \alpha_{jj'} \sum_{i' \in I(j',c')} VA_{i'j'c',t} \quad (47)$$

$$DMA_{jct} = \sum_{c' \neq c} \tau_{cc',t}^{-\theta} \sum_{j' \neq j} \alpha_{jj'} \sum_{i' \in I(j',c')} Sale_{i'j'c',t} \quad (48)$$

where i , j , c and t are the indices of firm, industry, city, and year, respectively.

$VA_{i'j'c',t}$ and $Sale_{i'j'c',t}$ represent the industrial value added and sales output value of firm i' in industry j' in destination city c' , respectively. $\alpha_{jj'}$ represents the input-output coefficient between industries j and j' , which is calculated based on the 2002 China Input-Output Table. τ is the index of the railway transportation cost per ton of goods shipped. θ represents the price elasticity of trade, which measures the substitution between the outputs of different cities. Following Donaldson (2018), we set the elasticity value to 3.8.

The specification of the cost of the railway transport follows the method of Baum-Snow et al. (2016), which is:

$$\tau_{cc',t} = 1 + p_t (d_{cc',t})^\sigma, \quad (49)$$

where p_t is the average annual freight rate. The average freight rate is obtained by dividing the total revenue of railway transportation, deflated by the production price index, by the total freight volume in tons. The variable $d_{cc',t}$ is the shortest railway distance between two cities

which is constructed by Gao and Tang (2024). σ is a parameter that captures the concave relationship between shipping cost and railway distance, set to 0.8 (Baum-Snow et al., 2016).

To address the endogeneity of MA, we construct two instrumental variables (IVs) rooted in exogenous geographic and historical factors. These IVs rely on two artificial transportation networks constructed by Gao and Tang (2024), which, in turn, adapts the core approaches of Faber (2014) and Banerjee et al. (2020) to build counterfactual infrastructure links.¹²

We use these two artificial transportation networks as foundational maps to compute counterfactual market access indicators: counterfactual UMA and counterfactual DMA. Thus, for each of UMA and DMA, we obtain a pair of counterfactual market access indices (one derived from the least-cost path network, the other from the straight-line network). These counterfactual indices serve as our IVs.

3.4 Data Description

After merging market access indices with firm-level data using the city code as the matching variable, we remove 1% of the observations that report extreme values of UMA and DMA at both ends to mitigate the influence of outliers on estimation. The descriptive statistics for the main variables are shown in Table 1. The average DVAR in the sample is 78.99% and exhibits substantial variation. The average age of the firms in the initial period is 1.735. Our sample is predominantly composed of private and foreign firms, as only 8.2% of the firm-year observations are SOEs. Export processing is common, with 41.7% of the observations reporting participation in processing.

Figure 3 shows that the average DVAR increased steadily during the period. When separating firms by their mode of participation in the trade, as shown in Figure 4, firms in the ordinary trade regime report the highest average DVAR at 90.94%, while firms in processing trade regime report the lowest average DVAR at 31.79%. Firms in the mixed-trade regime, that

¹²Gao and Tang (2024) first identify 36 key cities that must be connected. These cities include the 27 provincial capitals in mainland China (which are prefecture-level or vice-provincial cities), the four municipalities directly under the control of the Central Government (Beijing, Shanghai, Tianjin and Chongqing), and five additional key cities (Dalian, Ningbo, Qingdao, Shenzhen, and Xiamen). Although these five cities have the status of vice-provincial cities, they are not provincial capitals. To connect these 36 key cities, Gao and Tang (2024) construct two artificial transportation networks. The first follows the methodology of Faber (2014) to use the ArcGIS algorithm to determine the least-cost path between two cities. The second follows Banerjee et al. (2020) to construct an alternative artificial network by connecting the 36 key cities with straight-line segments. Since the locations of these cities are historically determined, any city at the prefecture level located near these artificial networks is more likely to integrate into the actual railway system due to exogenous geographic factors.

is, firms engaged in both ordinary trade and processing trade, exhibit an intermediate average DVAR of 67.81%.

To further explore DVAR differences across firm characteristics, Figure 5 tracks trends in DVAR by ownership type. Private firms have a higher DVAR than state-owned enterprises (SOEs) and foreign firms, maintaining a consistently high level. SOEs rank second, with their DVAR generally above 70.00%. Although foreign firms have the lowest DVAR, it shows gradual growth, suggesting that foreign firms source more intermediate inputs from domestic suppliers over time.

Figure 6 illustrates a considerable increase in both upstream market access (UMA) and downstream market access (DMA) over the sample period. Turning to regional disparities, Figure 7 highlights substantial differences in market access across China’s eastern, central, and western regions: despite the recent rapid expansion of the railway network in central and western China, their UMA and DMA levels remain significantly lower than those in the eastern region.

For the growth dynamics of market access by region, Figure 8 and Figure 9 respectively show the spatial distribution of UMA and DMA across major regions in the first and last years of the sample period. A notable pattern emerges here: while China’s western and central regions have registered more new railway connections, the eastern region has seen more widespread and robust growth in both UMA and DMA.

4 Regression Results

4.1 Benchmark Regressions Results for DVAR

We report the results of the OLS regressions in Table 2 which indicate that UMA is positively correlated with DVAR. According to column (2), which includes EUMA as a regressor based on equation (44), the firm’s DVAR is estimated to be 0.00203% higher if upstream market access increases by 1%. However, the estimate is not statistically significant.

In Table 3, we estimate the benchmark regression model based on the control function approach, using each of the two IVs separately, as well as both IVs. First, we use the two IVs individually and report the results in columns (1) and (2). The K-P rk Wald F statistic and the K-P rk LM statistic suggest that both IVs pass the weak identification and under-identification

tests. The first stage regression results show that the IVs are significantly related to the UMA index. In the second-stage regressions, the coefficients of the UMA index are positive and significant. When we employ both IVs together, the results shown in column (3) are robust, and the Hansen J statistic indicates that the regression passes the over-identification test.

Following Angrist and Kolesár (2024), who recommend a single-variable just-identified instrumental variables estimator, we rely on the first IV alone for identification in subsequent regressions as it incorporates more geographic information. Based on the results in column (1), the DVAR of the firms is estimated to increase by about 0.01333% when upstream market access increases by 1%, which is substantially higher than the estimate of OLS. Thus, the key prediction of Proposition 1, i.e., the improvement of market access increases firms' DVAR, is supported.

To evaluate the degree to which UMA explains the variations in DVAR, we adopt the methodology of Korovkin and Makarin (2023) and compute the following metric:

$$\Delta DVAR \text{ explained} = 100 \times \frac{[\ln(UMA_{p75}) - \ln(UMA_{p25})] \times |\beta_{UMA}|}{DVAR_{p75} - DVAR_{p25}}, \quad (50)$$

where $\ln(UMA_{p75})$ and $\ln(UMA_{p25})$ denote the 75th and 25th percentiles of the natural logarithm of UMA, respectively, while β_{UMA} represents the estimated coefficient of UMA in the DVAR regression. The above metric quantifies the proportion of differences in DVAR between the 25th and 75th percentile that can be attributed to variation in UMA over the same percentiles. Using the results of columns (1) in Table 3, we find that $\Delta DVAR \text{ explained} = 11.57\%$. Details of the calculation are contained in Table 4. Therefore, after controlling for firm-level and city-level factors, UMA explains a significant share of the observed increase in DVAR for Chinese firms' exports which indicates a change in the composition of intermediate goods toward domestic sources.

4.2 The Impact Mechanism

Choices of intermediate inputs are the core channel in our model through which UMA affects the composition of intermediate goods and the scale of production in vertical linkages. Proposition 2 states that an increase in domestic upstream market access leads to an increase in both the total variety of intermediate goods used by firms in production and the variety of imported intermediate goods. However, the share of imported intermediates in the total intermediate

goods is predicted to decrease. Proposition 3 predicts that an increase in access to the domestic upstream market increases the quantity of each intermediate good imported by firms.

To test Propositions 2 and 3, we obtain information about imported and domestic intermediate goods in the following steps. First, we extract the first six digits of the HS codes associated with each firm from the China Customs Trade Statistics database. These codes are then converted to the 1996 version of the HS codes using the concordance table provided by the United Nations¹³. Second, we map the HS product codes to the product codes of broad economic categories (BEC) and apply the classification strategy outlined in Upward et al. (2013) to identify intermediate goods among imported products. This process allows us to obtain for each firm both the number of intermediate goods imported and the import volume for each intermediate good. Finally, to measure the types of intermediate goods imported by firms, we follow the approach in Feenstra (1994) and Broda and Weinstein (2006), grouping products classified under the same six-digit HS code and originating from the same country (or region) as a category of products. Since directly measuring the variety of domestic intermediate goods is not feasible, we use the variety of intermediate goods exported by an ordinary trade firm as a proxy for the intermediate goods it uses. As this proxy approach limits the sample to the set of firms engaged in ordinary trade, the results should be interpreted with caution.

The regression results in columns (1) and (2) of Table 5 show that a 1% increase in UMA leads to a 0.059% increase in the number of imported intermediate inputs, and the average import value per intermediate input increases by 0.057%. Column (3) reveals that an increase 1% in UMA increases the variety of domestic intermediate goods by 0.066%. In column (4), a 1% increase in UMA is estimated to reduce the share of imported intermediate inputs in the firms' output by 0.006%.

In general, the results support Propositions 2 and 3, thus validating the mechanism outlined in our theory. As the number of imported intermediate inputs and the average import value per input increase, foreign producers of intermediate goods also benefit from improvements in China's infrastructure. Meanwhile, the share of imported inputs in total input costs decreases, contributing to an increase in DVAR.

¹³The concordance table is available at the link <https://unstats.un.org/unsd/classifications/Econ>.

4.3 Gains for Downstream and Upstream Firms Along Vertical Linkages

In Table 6, we test the prediction of Proposition 4, namely, a positive scale effect where a reduction in domestic trade costs increases the revenue of downstream firms. Although for simplification, the downstream firms in the model combine intermediate products to produce final goods and make zero profit, in the regressions we examine two additional outcome variables that are commonly used: value added and profit. The regression coefficients in columns (1) and (2) reveal that both the revenue derived from the core business and value added a firm respond positively to UMA. Following a 1% increase in UMA, revenue and value added increase by 0.032% and 0.059%, respectively. Although the estimated coefficient on UMA for profit is positive, it is highly insignificant, as the magnitude of the coefficient is small relative to the standard error. In general, Proposition 4 is supported.

To further analyze the gains from trade along vertical linkages, we examine how DMA affects upstream firms. The results presented in Table 7 indicate a clear pattern: an increase in DMA boosts the revenue from the core business, value added and profit of the upstream firms. The elasticities are 0.011, 0.011 and 0.016, respectively. These results further support the theoretical prediction that upstream firms benefit from market access.

In general, the analysis in this subsection clearly shows that improvements in market access resulting from the expansion of the railway network create trade gains for both downstream firms and upstream firms along vertical linkages.

When comparing the impact of market access on upstream and downstream firms, the similarities lie in the significant increases in both value added and main business revenue for both types of firms. The difference, however, is that upstream firms experience an increase in profits while downstream companies do not. This could be because a rise in UMA would lead to the entry of more downstream firms, which in turn intensifies competition and potentially offsets the cost savings brought about by the increase in UMA. In addition, the estimated market access elasticities of revenue and value added are larger for downstream firms than for upstream firms.

Therefore, the effect of increased market access on upstream and downstream firms exhibits a certain degree of asymmetry.

4.4 Further Evidence of Vertical Linkages: The Positive Role of Upstream Industries' Technology Level

If the core mechanism behind the increase in DVAR is improved access to upstream domestic inputs, as hypothesized in the current study, we expect the positive effect of UMA on DVAR to be stronger when upstream supply embodies a higher level of technology. To test this idea and further validate the role of vertical linkages, we construct a measure of the upstream industries' technology level. Specifically, for firms in industry j , the upstream industries' technology level (UIT) is defined as

$$UIT_j = \sum_k d_{jk} \cdot patent_k \quad (51)$$

where d_{jk} represents the direct consumption coefficient between industry j (downstream) and industry k (upstream), obtained from the China Input-Output Table in 2002 (122 sectors). $patent_k$ denotes the total number of patent applications filed by firms in industry k in 2002, reflecting the innovation capacity and technological sophistication of upstream suppliers.

Using the computed UIT_j , we split industries into two groups based on the median value of UIT . Column (1) of Table 8 reports results for firms in industries with above-median levels of UIT_j : the effect of UMA on DVAR is positive and statistically significant. In contrast, column (2) shows that for firms in industries with below-median level of UIT_j , the UMA coefficient is close to zero and highly insignificant.

To confirm robustness, we redefine $patent_k$ in equation (51) as the total number of citations of patents owned by firms in industry k in 2002, a citation-based measure of technology. Columns (3) and (4) of Table 8 replicate the split-sample analysis using this alternative UIT_j . The results are consistent: UMA significantly boosts DVAR only for firms in industries whose upstream industries' technology levels are above median.

These findings provide further evidence for our core mechanism: improved access to domestic intermediate goods enhances downstream firms' DVAR, but this effect is dependent on the level of technology of upstream industries. Firms in industries with suppliers in technologically advanced industries benefit disproportionately from increased market access.

4.5 Spatial Consideration: Interaction between Market Access and Coastal Location and Agglomeration

In this subsection, we analyze whether market access interacts with two important drivers of spatial concentration: coastal location advantages and agglomeration economies.

Proposition 5 predicts that the effect of UMA on DVAR is larger for the final good firms in port cities. To test this prediction, we include in the regression an interaction term between UMA and an indicator for port cities¹⁴. The results in column (1) of Table 9 show that the total effect of UMA on DVAR in port cities is 2.422 (sum of the UMA main effect [-0.381] and $UMA \times \text{port}$ interaction [2.803])—far larger than the benchmark UMA effect of 1.333 in Table 3. In contrast, the main effect of UMA (-0.381) is insignificant for inland cities, indicating that the gains in DVAR associated with UMA are concentrated in port cities.

Regarding agglomeration, Proposition 6 conjectures that a higher degree of agglomeration amplifies the effect of the railway infrastructure on DVAR under the assumption that access to intermediates and agglomeration are complements in strengthening the advantage of home intermediate good firms. To measure the industrial agglomeration for each firm’s industry in a city, we follow Koo (2007) and define agglomeration as the ratio of the employment density of industry j in city c relative to the national employment of industry j

$$agglomeration_{jc} = \frac{\text{employment of industry } j \text{ in city } c}{\text{national employment of industry } j \times \text{land area of city } c}.$$

Column (2) of Table 9 reports the regression that includes the interaction term between industrial agglomeration and UMA. As the coefficient on the interaction between agglomeration ($\ln(agg)$) and UMA ($\ln(UMA)$) is significant at the level 10% and positive, the result weakly supports the prediction of Proposition 6 that industrial agglomeration indeed strengthens the positive effect of UMA on DVAR.

In Table 10, we examine whether UMA affects value added, profit and revenue differently in port cities and in cities with a higher degree of agglomeration. In column (2), the large and significant coefficient of the interaction term between the UMA and port cities (coefficient = 0.083) contrasts sharply with the small and insignificant coefficient of the standalone UMA variable (coefficient = 0.008). This difference in coefficients suggests that UMA mainly boosts

¹⁴The port cities include: Dalian, Yingkou, Qinhuangdao, Tianjin, Yantai, Qingdao, Rizhao, Shanghai, Lianyungang, Nantong, Suzhou, Zhenjiang, Nanjing, Ningbo, Zhoushan, Wenzhou, Fuzhou, Xiamen, Shantou, Shenzhen, Guangzhou, Zhuhai, Zhanjiang, Fangchenggang, and Haikou.

the value added of final good firms in port cities but not in non-port cities. Meanwhile, because the interaction term between UMA and agglomeration in column (5) is only marginally significant (coefficient = 0.005; p-value = 0.130), there is weak evidence that the value added increased more in cities with a higher degree of agglomeration. In columns (1), (3), (4) and (6), there appear to be no interaction effects between UMA and port location or agglomeration on revenue and profit.

Taking the results from Table 9 and Table 10 together, we can see that UMA has a positive and large interaction effect with port location: following an increase in UMA, there are large increases in the DVAR and value added of downstream firms in port cities, but not in non-port cities. In comparison, the interaction effect between UMA and agglomeration is also positive but moderate in magnitude. Therefore, although railway infrastructure improves market access in both port and non-port cities, from the perspective of final good firms, most of the gains from trade accrue to port cities.

5 Robustness Checks

5.1 Accounting for Potential Confounding Factors

Our results remain robust when we explicitly control for a number of factors that may have affected firms' DVAR in the 2000's decade.

The first is trade liberalization, which increases the availability of imported intermediate goods. To isolate the effect of UMA from trade policy changes, we follow Lu et al. (2017) to include interaction terms between year dummies and industry-specific trade policy indicators in the regression. These indicators, which include industry-level output tariffs, input tariffs, and export tariffs, capture the time-varying effects of trade liberalization on different industries. The results of this specification, reported in column (1) of Table 11, show that the effects of UMA remain positive and significant.

A second potential confounder is the unique administrative and economic status of China's four directly administered municipalities (Beijing, Chongqing, Shanghai, and Tianjin). These municipalities have a provincial administrative rank (higher than prefecture-level cities) and receive disproportionate weight in national economic planning, including prioritized investments in transportation infrastructure. To rule out the possibility that our results are driven by these

exceptional regions, we exclude firms located in these four municipalities and re-estimate the benchmark regression. Column (2) of Table 11 presents these results, which remain consistent with our benchmark findings.

The third potential confounding factor is unobserved industry- or region-specific trends that affect the DVAR of related firms. For example, increase in policy preference to develop regions with more ethnic minority groups that can lead to both more railway construction and sourcing from local producers (hence boosting DVAR). To address such concerns, we sequentially add industry-time trends and industry-year fixed effects to the benchmark regression. The results are reported in columns (3) and (4) of Table 11, both of which are consistent with the benchmark results.

Furthermore, we also use long-difference regressions to mitigate the potential bias caused by unobserved trends. The results presented in Table 12 confirm that after removing the effects of short-term fluctuations or unobserved trends, we find a positive long-term impact of UMA on the DVAR of firms. Thus, our results are generally robust to the influence of potential confounding factors.

5.2 Alternative Measurement of DVAR and Market Access

In this subsection, we first show that our results are robust to alternative measure of dependent variable. As discussed in Section 3, our calculation of DVAR modifies the method proposed by Upward et al. (2013). To verify robustness, we re-estimate the regression using DVAR calculated with the original method of Upward et al. (2013). Column (1) of Table 13 reports these results, which are consistent with the benchmark regression results in column (1) of Table 3.

We also examine whether the results are affected by alternative parameters in the construction of the UMA. In columns (2) to (6), we consider alternative parameter combinations in the construction of the UMA index. Specifically, we vary the values of θ (trade elasticity) and σ (freight rate elasticity) when calculating market access, yielding five alternative UMA measures¹⁵. Lastly, we construct UMA using PPI-deflated freight rates (instead of nominal freight rates) to account for price-level differences between regions. The results of this specification

¹⁵The parameter combinations are as follows: UMA_{t2} uses $\theta = 4.1$ and $\sigma = 0.8$; UMA_{t3} uses $\theta = 8.3$ and $\sigma = 0.8$; UMA_{t4} uses $\theta = 6$ and $\sigma = 0.8$; UMA_{s2} uses $\theta = 3.8$ and $\sigma = 0.6$; UMA_{s3} uses $\theta = 3.8$ and $\sigma = 1$.

are reported in column (7) of Table 13.

Across all regressions based on alternative measures, reported in Table 13, the coefficient on UMA remains positive and statistically significant. Their magnitudes are closely aligned with the benchmark regression.

Lastly, in Table 14, we present regressions with lags of UMA. The coefficients indicate that UMA affects the DVAR mainly in the same period and there are no significant lag effects. This is reasonable because the planned expansion in the railway network is public information, and firms would rationally take into account the reduction in domestic trade costs in their decision making.

6 Conclusion

The current study examines the relation between two important features of China’s development trajectory: massive investment in transportation infrastructure and a tendency of exporter firms to source more intermediate inputs domestically, the latter being an uncommon trend during trade liberalization. Specifically, we investigate how the expansion of China’s domestic railway network affects sourcing choices of exporter firms as measured by the domestic value added ratio (DVAR), and further examine whether it generates trade gains for upstream firms (both domestic and foreign) from a vertical trade perspective.

To achieve this goal, we developed a Melitz model in which final good firms endogenously choose varieties of domestic and foreign intermediates. The expansion of the railway improves access to domestic intermediates by reducing domestic trade costs, generating two key mechanisms. The first is a composition effect; as domestic trade costs fall more for domestic than foreign intermediates, downstream firms shift sourcing to domestic inputs, raising their DVAR. The second is scale effect; i.e. expanded input access boosts the variety and quantity of intermediates used, driving production growth. The model predicts: (1) DVAR rises via the composition effect; (2) upstream firms (including foreign suppliers) and downstream firms gain from scale effect-driven revenue growth; (3) DVAR growth is concentrated in port and agglomerated cities.

Using 2000–2007 firm and customs data, with a control function identification strategy, our empirical analysis confirms the model’s main predictions. Consistent with the composition

effect, upstream market access (UMA) significantly raises exporters' DVAR, as it explains 11.57% of interquartile variation in DVAR. Revenue and value added of exporters increases with UMA, confirming the scale effect. Downstream market access (DMA) improves performance among domestic and foreign upstream firms; foreign suppliers gain from increased varieties and quantities of Chinese imports of intermediates, reflecting the dominant scale effect. Finally, UMA improves DVAR and value added only for exporter firms in port cities, confirming spatial concentration of trade gains.

The findings carry important policy implications: this research provides new insights into how infrastructure shapes trade gains in global value chains, thereby helping policymakers better understand the economic benefits of domestic infrastructure. This study certainly has limitations. First, given that our sample period ends in 2007, future research could extend the data to capture post-2007 railway expansions. Second, while market access explains 11.57% of the interquartile variation in DVAR, a substantial fraction of the increase in China's DVAR remains unexplained. Future research should explore the role of other factors in determining DVAR.

Declaration of generative AI technologies in the writing process

During the preparation of this work, the authors used GPT 4o developed by OpenAI in order to improve readability and language. After using this tool/service, the authors reviewed and edited the content as needed and assume full responsibility for the content of the publication.

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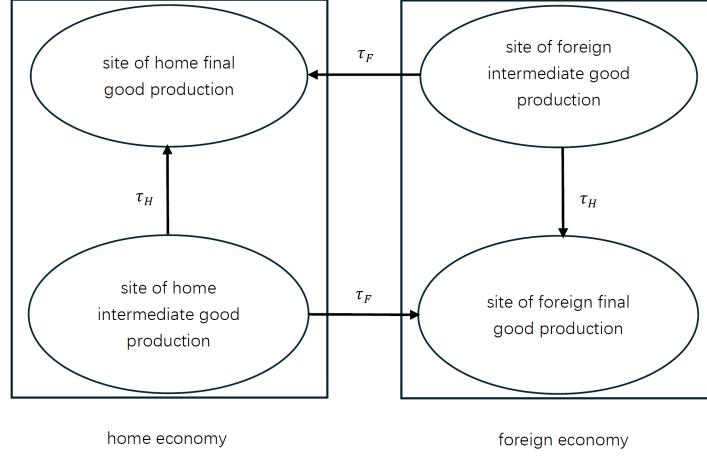
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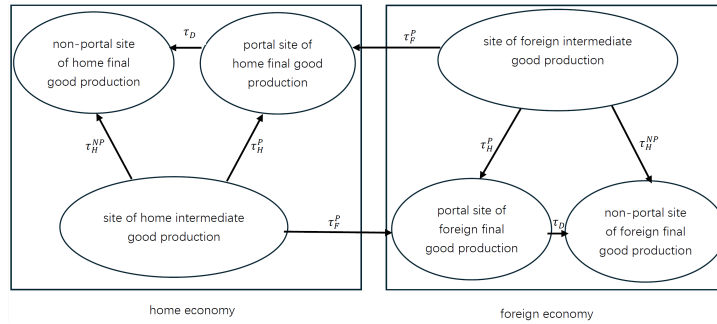
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Figure 1: Schematic Illustration of Spatial Structure in the Basic Model



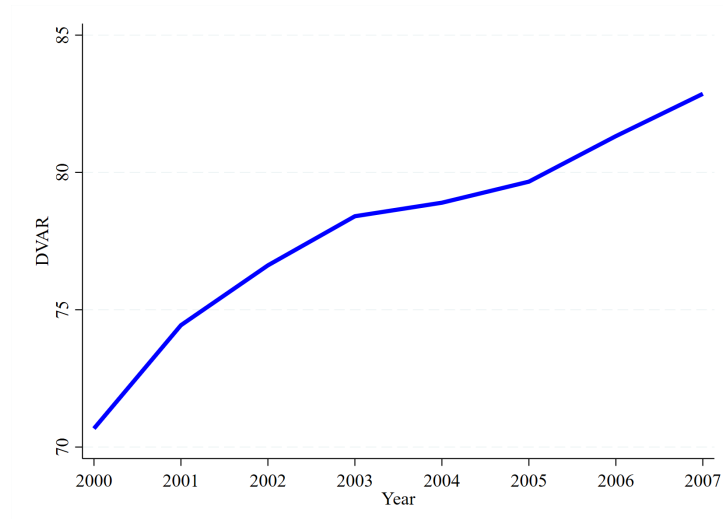
Note: This figure illustrates the geographic structure in our benchmark theoretical setup.

Figure 2: Schematic Illustration of Spatial Structure in the Extended Model



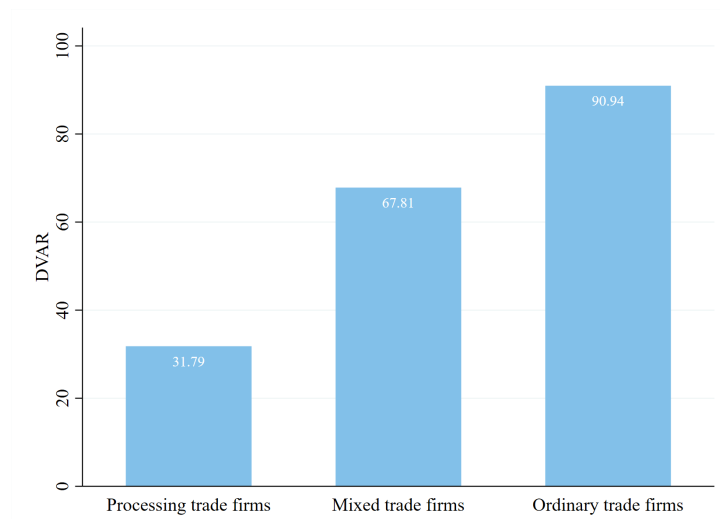
Note: This figure illustrates the geographic structure in our extended theoretical setup in which we introduce port and non-port cities to the model.

Figure 3: Average Firm-level Domestic value added Ratio (DVAR)



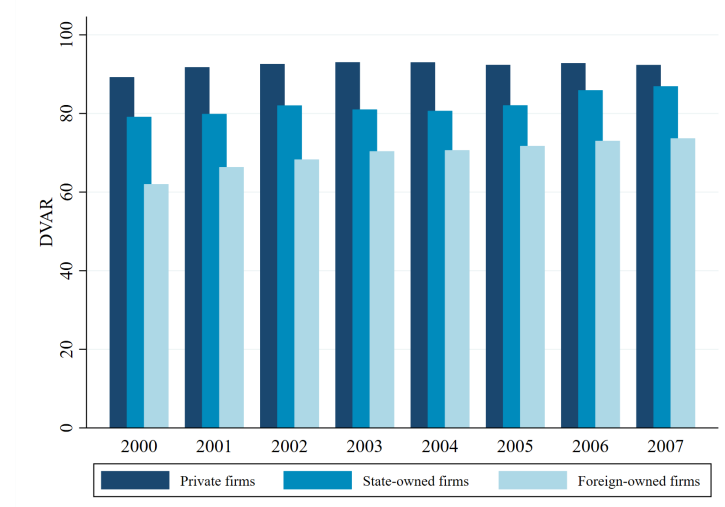
Note: This figure presents the trend in the average DVAR of Chinese firms over the period 2000–2007. This index was computed by the authors following the methodology proposed in Kee and Tang (2016). The underlying data were constructed by merging firm-level data from the Annual Survey of Industrial Firms and the China Customs Trade Database.

Figure 4: Average Firm-level Domestic value added Ratio (DVAR) by Trade Mode



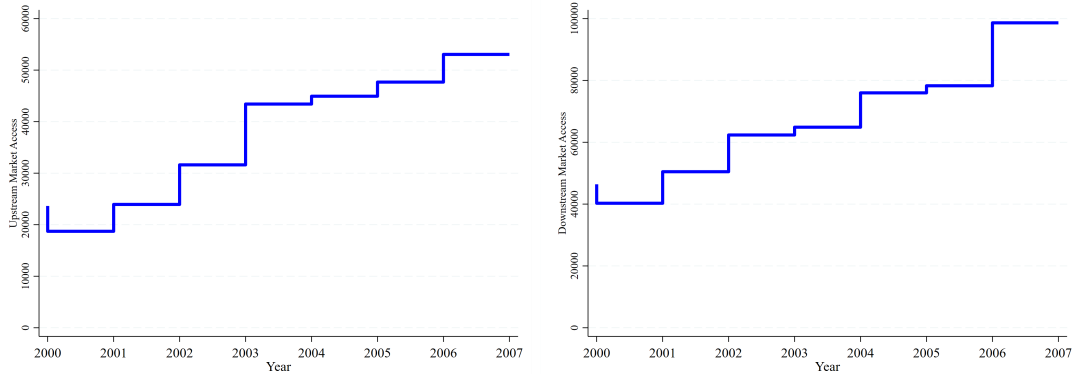
Note: This figure presents the average DVAR of Chinese firms by trade mode for the entire sample period. Firms' trade modes include processing trade, mixed trade, and ordinary trade. The trade mode to which a firm belongs is directly identified from customs data.

Figure 5: Average Firm-level Domestic value added Ratio (DVAR) by Ownership Types



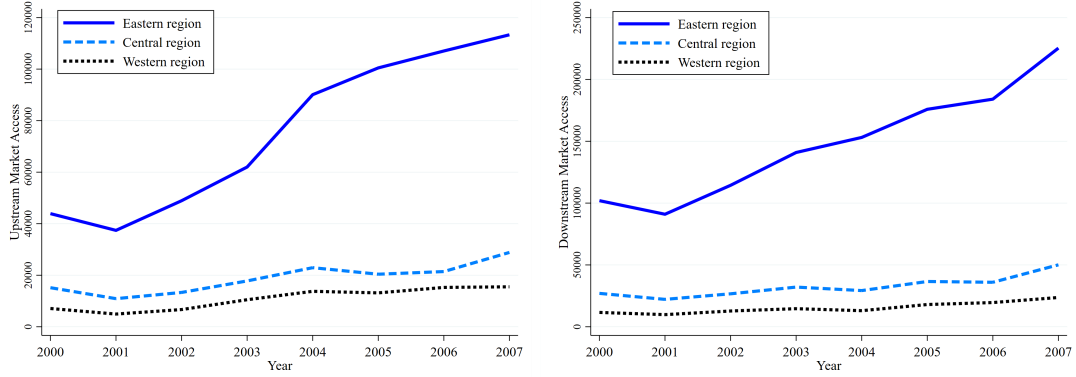
Note: This figure presents the trend of average DVAR of Chinese firms by ownership type over the period 2000–2007. The ownership type of a firm is directly identified from customs data.

Figure 6: Average Upstream and Downstream Market Access Indices



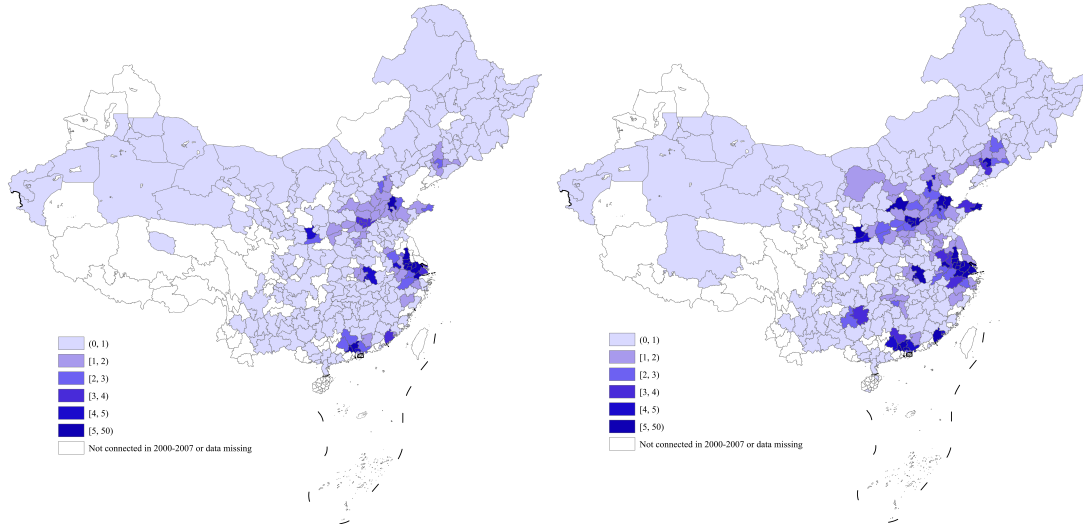
Note: The left and right panels present the trends in the average upstream market access (UMA) and downstream market access (DMA) of Chinese firms over the period 2000–2007, respectively. These indices were calculated by the authors using the method proposed in Donaldson and Hornbeck (2016), applied to railway network data constructed by Gao and Tang (2024). The underlying data are sourced from the *China Railway Yearbook*, *Chinese Research Data Services Platform*, and *China Urban Statistical Yearbook*.

Figure 7: Average Upstream and Downstream Market Access by Region



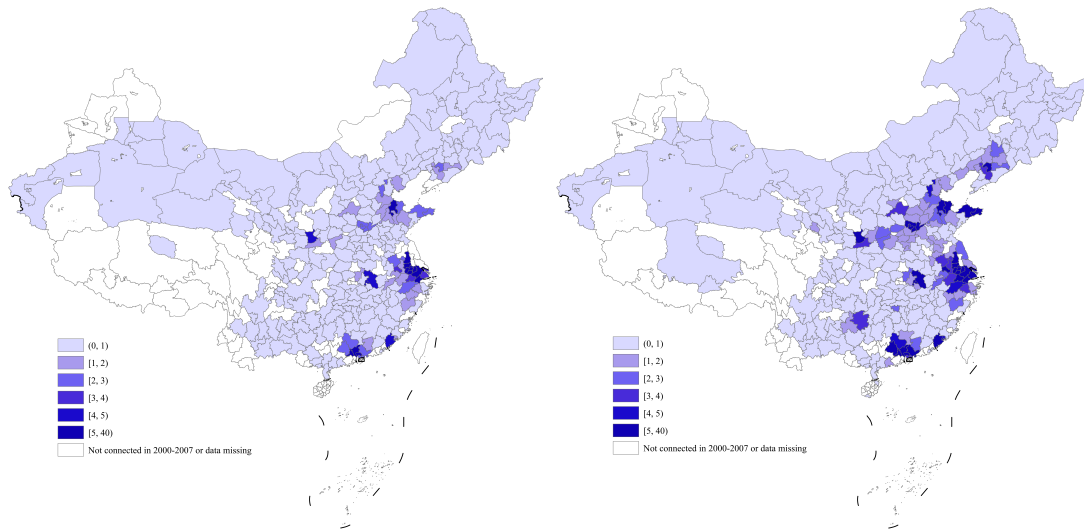
Note: The left and right panels present the trends in average upstream market access (UMA) and downstream market access (DMA) across different regions over the period 2000–2007, respectively. The solid line, dashed line, and dotted line denote the eastern, central, and western regions, respectively. The eastern region includes the following province-level administrative units: Beijing, Tianjin, Hebei, Liaoning, Shanghai, Jiangsu, Zhejiang, Fujian, Shandong, Guangdong, and Hainan. The central region includes: Shanxi, Inner Mongolia, Jilin, Heilongjiang, Anhui, Jiangxi, Henan, Hubei, Hunan, and Guangxi. The western region includes: Chongqing, Sichuan, Guizhou, Yunnan, Tibet, Shaanxi, Gansu, Qinghai, Ningxia, and Xinjiang.

Figure 8: The Spatial Distribution of Upstream Market Access in 2000 and 2007



Note: The left and right panels present the spatial distribution of upstream market access (UMA) in 2000 and 2007, respectively. The values in the left and right maps represent the ratio of each city's UMA in 2000 and 2007 to the mean UMA level in 2000, respectively. Blank cities correspond to those unconnected to the railway network in 2000 (left map) or 2007 (right map), or those with missing data. Map lines delineate the study areas and do not necessarily reflect officially recognized national boundaries. The darker the color in the maps, the higher the UMA value.

Figure 9: The Spatial Distribution of Downstream Market Access in 2000 and 2007



Note: The left and right panels present the spatial distribution of downstream market access (DMA) in 2000 and 2007, respectively. The values in the left and right maps represent the ratio of each city's DMA in 2000 and 2007 to the mean DMA level in 2000, respectively. Blank cities correspond to those unconnected to the railway network in 2000 (left map) or 2007 (right map), or those with missing data. Map lines delineate the study areas and do not necessarily reflect officially recognized national boundaries. The darker the color in the maps, the higher the DMA value.

Table 1: Summary Statistics

	Observations	Mean	Min	Max	S.D.
Firm-level variables					
DVAR(%)	189927	78.99	0.000	100.00	30.02
processing	189927	0.417	0	1	0.493
SOE	189927	0.082	0	1	0.275
HHI (initial period)	50436	0.062	0.002	1.000	0.088
age (initial period)	50436	1.735	0	4.043	0.774
City-industry-level variables					
ln(UMA)	30023	10.04	7.051	13.89	1.520
IV 1 (artificial ln(UMA))	30023	9.174	6.370	12.95	1.444
IV 2 (artificial ln(UMA))	30023	9.271	6.512	12.81	1.389
ln(EUMA) (initial period)	5817	9.930	5.075	17.41	1.432
ln(DMA)	78265	9.422	4.507	15.31	2.169
IV 1 (artificial ln(DMA))	78265	8.609	3.766	14.33	2.086
IV 2 (artificial ln(DMA))	78265	8.687	-0.425	15.58	2.120
ln(EDMA) (initial period)	12198	9.449	-0.321	15.57	2.158
City-level variables					
ln _{dist} (initial period)	230	6.309	4.269	8.082	0.706
lnGDP (initial period)	230	14.72	12.10	17.63	0.894
ln _{wage} (initial period)	230	8.972	8.305	10.05	0.269
investment_rate (initial period)	230	0.226	0.056	0.776	0.103
road_density (initial period)	230	0.368	0.059	1.708	0.195

Notes: [1] *DVAR* is the firm-level Domestic Value Added Ratio (DVAR) in exports. *processing* and *SOE* represent the indicator for firms engaged in processing trade and the indicator for state-owned enterprises, respectively. *HHI* and *age* are the Herfindahl–Hirschman Index of the four-digit China Industry Classification industry to which a firm belongs and the firm’s age measured as the log of firm age plus 1, respectively. [2] *ln(UMA)* (respectively, *ln(DMA)*) is the log of upstream (respectively, downstream) market access index. *IV1* is the counterfactual UMA (or DMA) that is constructed by the artificial transportation network connected by the least costly paths, and *IV2* is the counterfactual UMA (or DMA) that is constructed by the artificial transportation network connected by straight lines. *ln(EUMA)* (respectively, *ln(EDMA)*) is the log of expected upstream (respectively, downstream) market access index. [3] *ln_{dist}*, *lnGDP*, *ln_{wage}*, *investment_rate*, and *road_density* represent the log of rail transport distance from the city to the nearest sea port, the log of the GDP of the city, the log of the average wage of the city, the ratio of fixed investment to GDP in the city, and the ratio of the total mileage of the road to the land area in the city. [4] For all variables with “initial period” indicated in parentheses, the numbers are the descriptive statistics for the initial period (2000).

Table 2: Upstream Market Access and DVAR (OLS)

Dependent Variable:	<i>DVAR</i>	
	(1)	(2)
$\ln(UMA)$	0.264 (0.342)	0.203 (0.330)
Dependent Variable's Mean	78.99	78.99
$\ln(EUMA_{jc}) \times \eta_t$	No	Yes
$X_{ijc} \times \eta_t$	Yes	Yes
$Z_c \times \eta_t$	Yes	Yes
Time-varying firm controls	Yes	Yes
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Observations	189,927	189,927
Adjusted R^2	0.763	0.763

Notes: [1] The dependent variable is the firms' DVAR. $\ln(UMA)$ is the log of upstream market access index. $\ln(EUMA)$ is the log of expected upstream market access index. [2] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. Standard errors are clustered at the city-industry level and shown in parenthesis.

Table 3: Upstream Market Access and DVAR (2SLS)

Dependent Variable:	Panel A. Second stage		
	<i>DVAR</i>		
	(1)	(2)	(3)
$\ln(UMA)$	1.333** (0.519)	1.149** (0.511)	1.236** (0.510)
Dependent Variable's Mean	78.99	78.99	78.99
$\ln(EUMA_{jc}) \times \eta_t$	Yes	Yes	Yes
$X_{ijc} \times \eta_t$	Yes	Yes	Yes
$Z_c \times \eta_t$	Yes	Yes	Yes
Time-varying firm controls	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	189,927	189,927	189,927
Dependent Variable:	Panel B. First stage		
	$\ln(UMA)$		
	(1)	(2)	(3)
<i>IV1</i>	0.691*** (0.017)		0.334*** (0.064)
<i>IV2</i>		0.706*** (0.018)	0.375*** (0.065)
Dependent Variable's Mean	10.792	10.792	10.792
$\ln(EUMA_{jc}) \times \eta_t$	Yes	Yes	Yes
$X_{ijc} \times \eta_t$	Yes	Yes	Yes
$Z_c \times \eta_t$	Yes	Yes	Yes
Time-varying firm controls	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
K-P rk LM statistic	425.30	424.82	426.31
K-P rk Wald F statistic	1593.7	1611.0	823.93
Hansen J statistic	\	\	2.173
Observations	189,927	189,927	189,927

Notes: [1] The dependent variable in Panel A is the firms' DVAR. $\ln(UMA)$ is the log of upstream market access index. $\ln(EUMA)$ is the log of expected upstream market access index. The dependent variable in Panel B is the UMA index. [2] *IV1* is the counterfactual UMA that is constructed by the artificial transportation network connected by the least costly paths, and *IV2* is the counterfactual UMA that is constructed by the artificial transportation network connected by straight lines. [3] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. Standard errors are clustered at the city-industry level and shown in parenthesis.

Table 4: Explanatory Power of Upstream Market Access

Variables	P_{25}	P_{75}	$P_{75}-P_{25}$	β	Change explained
$\ln(UMA)$	9.772	11.98	2.208	1.333	11.57%
$DVAR$	72.11	97.56	25.45		

Note: The explanatory power statistic in the last column is calculated as equation 50. p_{25} and p_{75} represent the 25th and 75th percentiles of the corresponding variables in the first column. The estimates of β is from Column (1) in Table 3.

Table 5: Impact Mechanism: Varieties and Quantities of Intermediate Inputs (2SLS)

Dependent Variables:	N_F number of foreign inputs (1)	X_{jiF} quantity of foreign inputs (2)	N_H number of home inputs (3)	$share_F$ share of foreign inputs (4)
$\ln(UMA)$	0.059*** (0.019)	0.057** (0.025)	0.066*** (0.026)	-0.006* (0.004)
Dependent Variable's Mean	18.245	6.437	9.415	0.106
$\ln(EUMA_{jc}) \times \eta_t$	Yes	Yes	Yes	Yes
$X_{ijc} \times \eta_t$	Yes	Yes	Yes	Yes
$Z_c \times \eta_t$	Yes	Yes	Yes	Yes
Time-varying firm controls	Yes	Yes	Yes	Yes
Year fixed effects	Yes	No	Yes	Yes
Firm fixed effects	Yes	No	Yes	Yes
Firm-product fixed effects	No	Yes	No	No
Product-year fixed effects	No	Yes	No	No
Country-year fixed effects	No	Yes	No	No
K-P rk LM statistic	/	132.16	/	425.15
K-P rk Wald F statistic	/	1152.0	/	1592.1
Wald statistic	491.77	/	309.23	/
Observations	134,933	2,029,520	148,145	189,909

Notes: [1] The dependent variables in columns (1) to (3) are the number of imported intermediate goods used by a firm, the log of firm's import volume of an intermediate good, and number of domestic intermediate goods used by a firm which is measured as the variety of intermediate goods exported by ordinary trade firms. The dependent variable in Column (4) is the share of imported intermediate goods in the cost of all intermediates goods. [2] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. Standard errors are clustered at the city-industry level and shown in parenthesis.

Table 6: Upstream Market Access and Firms' Performance (2SLS)

Dependent Variables:	$\ln(revenue)$ (1)	$\ln(value\ added)$ (2)	$\ln(profit)$ (3)
$\ln(UMA)$	0.032** (0.016)	0.059*** (0.021)	0.002 (0.037)
Dependent Variable's Mean	10.767	9.370	7.474
$\ln(EUMA_{jc}) \times \eta_t$	Yes	Yes	Yes
$X_{ijc} \times \eta_t$	Yes	Yes	Yes
$Z_c \times \eta_t$	Yes	Yes	Yes
Time-varying firm controls	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
K-P rk LM statistic	425.27	425.30	417.09
K-P rk Wald F statistic	1592.8	1593.7	1518.2
Observations	189,922	189,927	150,531

Notes: [1] The dependent variables in columns (1) to (3) are the log of firms' revenue from main business, the log of firms' industrial added value, and the log of firms' profit. [2] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. Standard errors are clustered at the city-industry level and shown in parenthesis.

Table 7: Downstream Market Access and Firms' Performance (2SLS)

Dependent Variables:	$\ln(revenue)$ (1)	$\ln(value\ added)$ (2)	$\ln(profit)$ (3)
$\ln(DMA)$	0.011*** (0.002)	0.011*** (0.003)	0.016*** (0.005)
Dependent Variable's Mean	10.046	8.730	6.542
$\ln(EDMA_{jc}) \times \eta_t$	Yes	Yes	Yes
$X_{ijc} \times \eta_t$	Yes	Yes	Yes
$Z_c \times \eta_t$	Yes	Yes	Yes
Time-varying firm controls	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
K-P rk LM statistic	1118.9	1119.1	1014.7
K-P rk Wald F statistic	24556	24550	24763
Observations	952,880	952,997	779,541

Note: [1] The dependent variables in columns (1) to (3) are the log of firms' revenue from main business, the log of firms' industrial added value, and the log of firms' profit. [2] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. Standard errors are clustered at the city-industry level and shown in parenthesis.

Table 8: Interaction Effects of UMA and Upstream Industry's Technology Level (2SLS)

	<i>DVAR</i>			
Dependent Variable:	(1)	(2)	(3)	(4)
subsample	<i>tech level</i> > <i>median</i>	<i>tech level</i> ≤ <i>median</i>	<i>tech level</i> > <i>median</i>	<i>tech level</i> ≤ <i>median</i>
$\ln(UMA)$	2.801*** (1.015)	0.096 (0.781)	2.757*** (0.979)	-0.264 (0.754)
Dependent Variable's Mean	75.43	83.24	75.08	83.76
$\ln(EUMA_{jc}) \times \eta_t$	Yes	Yes	Yes	Yes
$X_{ijc} \times \eta_t$	Yes	Yes	Yes	Yes
$Z_c \times \eta_t$	Yes	Yes	Yes	Yes
Time-varying firm controls	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
K-P rk LM statistic	152.85	220.94	166.53	217.91
K-P rk Wald F statistic	406.92	723.96	453.51	729.92
Observations	102,030	86,377	103,230	85,235
Fisher's permutation test	2.705***		3.021***	

Notes: [1] The dependent variable is firms' DVAR in all columns. [2] The subsample used in column (1) (column (2), respectively) includes firms in industries for which the upstream industry's technology level is above-median (below-median, respectively), and the technology level is measured by the number of patent applications. [3] The subsample used in column (3) (column (4), respectively) includes firms in industries for which the upstream industry's technology level is above-median (below-median, respectively), and the technology level is measured by the number of patent citations. [4] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. Standard errors are clustered at the city-industry level and shown in parenthesis.

Table 9: Interaction Effects of UMA and Location and Agglomeration on DVAR (2SLS)

Dependent Variable:	<i>DVAR</i>	
	(1)	(2)
$\ln(UMA)$	-0.381 (0.601)	1.311** (0.518)
$\ln(UMA) \times port$	2.803*** (0.828)	
$\ln(UMA) \times \ln(agg)$		0.114* (0.068)
$\ln(agglomeration)$		-0.130 (0.094)
Dependent Variable's Mean	78.99	78.99
$\ln(EUMA_{jc}) \times \eta_t$	Yes	Yes
$X_{ijc} \times \eta_t$	Yes	Yes
$Z_c \times \eta_t$	Yes	Yes
Time-varying firm controls	Yes	Yes
Year fixed effects	Yes	Yes
Firm fixed effects	Yes	Yes
K-P rk LM statistic	490.23	460.10
K-P rk Wald F statistic	540.13	837.27
Observations	189,927	189,927

Notes: [1] The dependent variable in all columns is the firms' DVAR. The variable *port* is an indicator for firms located at a port, and the variable $\ln(agglomeration)$ represents city-industry level industrial agglomeration. [2] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. Standard errors are clustered at the city-industry level and shown in parenthesis.

Table 10: Interaction Effects of UMA and Location and Agglomeration on Revenue, Value added, and Profit (2SLS)

Dependent Variables:	$\ln(revenue)$ (1)	$\ln(value\ added)$ (2)	$\ln(profit)$ (3)	$\ln(revenue)$ (4)	$\ln(value\ added)$ (5)	$\ln(profit)$ (6)
$\ln(UMA)$	0.019 (0.023)	0.008 (0.030)	0.004 (0.054)	0.027* (0.016)	0.054*** (0.021)	-0.0004 (0.037)
$\ln(UMA) \times port$	0.021 (0.029)	0.083** (0.036)	-0.003 (0.063)			
$\ln(UMA) \times \ln(agg)$				0.003 (0.003)	0.005 (0.003)	-0.004 (0.006)
$\ln(agg)$				0.060*** (0.004)	0.060*** (0.005)	0.074*** (0.008)
Dependent Variable's Mean	10.767	9.370	7.474	10.767	9.370	7.474
Expected Market Access	Yes	Yes	Yes	Yes	Yes	Yes
$W_{ijc} \times \eta_t$	Yes	Yes	Yes	Yes	Yes	Yes
$Z_c \times \eta_t$	Yes	Yes	Yes	Yes	Yes	Yes
Time-varying firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
K-P rk LM statistic	490.30	490.23	444.03	460.10	460.08	449.95
K-P rk Wald F statistic	540.12	540.27	486.30	837.37	837.14	801.41
Observations	189,922	189,927	150,531	189,922	189,927	150,531

Note: [1] The dependent variables in columns (1) to (3) are the log of firms' revenue from main business, the log of firms' industrial added value, and the log of firms' profit. The dependent variables in columns (4) to (6) repeat those in columns (1) to (3). The variable *port* is an indicator for firms located at a port, and the variable $\ln(agg)$ represents city-industry level industrial agglomeration. [2] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. Standard errors are clustered at the city-industry level and shown in parenthesis.

Table 11: Robustness Checks: Potential Confounding Factors (2SLS)

Dependent Variable:	<i>DVAR</i>			
	(1)	(2)	(3)	(4)
$\ln(UMA)$	1.430*** (0.538)	1.277** (0.583)	2.573*** (0.734)	3.707*** (1.152)
Dependent Variable's Mean	78.99	80.65	78.99	78.99
$\ln(EUMA_{jc}) \times \eta_t$	Yes	Yes	Yes	Yes
$X_{ijc} \times \eta_t$	Yes	Yes	Yes	Yes
$Z_c \times \eta_t$	Yes	Yes	Yes	Yes
Time-varying firm controls	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Industry time trends	No	No	Yes	No
Industry-year fixed effects	No	No	No	Yes
Tariff reductions $\times$ year dummies	Yes	No	No	No
K-P rk LM statistic	443.59	460.75	475.62	378.56
K-P rk Wald F statistic	1580.8	1307.4	989.70	530.87
Observations	189,927	157,754	189,927	189,924

Notes: [1] The dependent variable is the firms' DVAR. [2] The regression in Column (1) includes the interaction between year dummies and industry levels of output tariff, input tariff, and export tariff as control variables. Column (2) excludes the firms from the four directly administered municipalities (Beijing, Chongqing, Shanghai, and Tianjin). Column (3) controls for the industry time trends. Column (4) controls for the industry-year fixed effects. [3] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. Standard errors are clustered at the city-industry level and shown in parentheses.

Table 12: Long Difference Analysis (2SLS)

Dependent Variable:	$\Delta DVAR$	
	(1)	(2)
$\Delta \ln(UMA)$	6.589*** (1.078)	6.552*** (1.082)
$\ln(EUMA_{jc})$		0.090 (0.149)
Dependent Variable's Mean	2.860	2.860
X_{ijc}	Yes	Yes
Z_c	Yes	Yes
K-P rk LM statistic	18.232	18.284
K-P rk Wald F statistic	512.55	524.23
Observations	81,821	81,821

Notes: [1] The dependent variable is the long difference of firms' DVAR, calculated as the end-of-period value minus the beginning-of-period value. [2] The variable $\Delta \ln(UMA)$ represents the difference between the end-of-period and beginning-of-period values of $\ln(UMA)$. The instrumental variable is the difference between the end-of-period and beginning-of-period values of $IV1$. [3] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. Standard errors are clustered at the city level and shown in parenthesis.

Table 13: Robustness Checks: Alternative Parameters in the Construction of MA (2SLS)

Dependent Variable:	DVAR						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\ln(UMA)$	1.522*** (0.494)						
$\ln(UMA)_{t2}$		1.316** (0.512)					
$\ln(UMA)_{t3}$			1.217*** (0.471)				
$\ln(UMA)_{t4}$				1.257** (0.488)			
$\ln(UMA)_{s2}$					1.553** (0.609)		
$\ln(UMA)_{s3}$						1.297** (0.505)	
$\ln(UMA)_{ppi}$							1.377** (0.538)
Dependent Variable's Mean	78.19	78.99	78.99	78.99	78.99	78.99	78.99
$\ln(EUMA_{jc}) \times \eta_t$	Yes	Yes	Yes	Yes	Yes	Yes	Yes
$X_{ijc} \times \eta_t$	Yes	Yes	Yes	Yes	Yes	Yes	Yes
$Z_c \times \eta_t$	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time-varying firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
K-P rk LM statistic	416.73	422.78	394.88	405.89	420.28	401.71	413.53
K-P rk Wald F statistic	1686.4	1555.1	1135.1	1293.3	1406.6	1286.7	1490.4
Observations	204,950	189,927	189,927	189,927	189,927	189,927	189,927

Notes: [1] The dependent variable is the firms' DVAR. [2] The independent variables in Columns (2)-(6) are logs of upstream market access constructed with alternative parameters of the price elasticity of trade, θ , and the concavity of shipping cost to railway distance, σ . These variables include $\ln(MA)_{t2}(\sigma = 0.8; \theta = 4.1)$, $\ln(MA)_{t3}(\sigma = 0.8; \theta = 8.3)$, $\ln(MA)_{t4}(\sigma = 0.8; \theta = 6)$, $\ln(MA)_{s2}(\sigma = 0.6; \theta = 3.8)$, and $\ln(MA)_{s3}(\sigma = 1; \theta = 3.8)$. $\ln(MA)_{ppi}$ has the same parameter values as $\ln(MA)$ but replaces nominal freight rate with freight price after PPI adjustment. [3] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. Standard errors are clustered at the city-industry level and shown in parenthesis.

Table 14: Robustness Checks: Lag Effects of Upstream Market Access (2SLS)

Dependent Variable:	<i>DVAR</i>	
	(1)	(2)
$\ln(UMA)$	1.444** (0.687)	1.533* (0.782)
$L.\ln(UMA)$	0.035 (0.375)	0.357 (0.503)
$L2.\ln(UMA)$		-0.453 (0.395)
Dependent Variable's Mean	80.28	80.89
$\ln(EUMA_{jc}) \times \eta_t$	Yes	Yes
$X_{ijc} \times \eta_t$	Yes	Yes
$Z_c \times \eta_t$	Yes	Yes
Time-varying firm controls	Yes	Yes
Year fixed effects	Yes	Yes
Firm fixed effects	Yes	Yes
K-P rk LM statistic	339.19	258.85
K-P rk Wald F statistic	833.69	823.58
Observations	113,004	68,320

Notes: [1] The dependent variable is the firms' DVAR. [2] The variables $L.\ln(UMA)$ and $L2.\ln(UMA)$ represent the one-period and two-period lagged value of $\ln(UMA)$. [3] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. Standard errors are clustered at the city-industry level and shown in parenthesis.

Appendix For Online Publication

A Optimal Quantity of Intermediate Goods

In Appendix A, we show how to obtain the optimal quantity of intermediate good i by solving the choice of final goods firm j , who minimizes the total cost

$$X_{ji} = \arg \min_{X_{ji}} \int_{i=1}^N P_i X_{ji} di,$$

subject to the constraint

$$X_j = \left(\int_{i=0}^N X_{ji}^\rho di \right)^{1/\rho}.$$

The first-order condition for the minimization can be written as

$$\frac{X_{ji}}{X_{jt}} = \left[\frac{P_t}{P_i} \right]^{\frac{1}{1-\rho}}. \quad (\text{A.1})$$

Substituting (A.1) into the constraint, we obtain

$$X_j = X_{ji} P_i^{\frac{1}{1-\rho}} \left[\int_{i=0}^N P_i^{\frac{\rho}{\rho-1}} di \right]^{\frac{1}{\rho}}. \quad (\text{A.2})$$

Then we can obtain the cost index of producing final good j , which satisfies the total cost equation

$$C_j X_j = \int_{i=1}^N P_i X_{ji} di.$$

Substituting (A.1) and (A.2) into the last line, we simplify to obtain

$$C_j = \left[\int_{i=0}^N P_i^{\frac{\rho}{\rho-1}} di \right]^{\frac{\rho-1}{\rho}}. \quad (\text{A.3})$$

Substituting (A.3) into (A.2) yields

$$X_{ji} = X_j \left[\frac{P_i}{C_j} \right]^{\frac{1}{\rho}-1}. \quad (\text{A.4})$$

Using the production function in equation (5), we can obtain

$$C_j X_j = P_j Q_j. \quad (\text{A.5})$$

Combining (A.5) and (A.4), we obtain the solution for the optimal quantity of X_{ji}

$$X_{ji} = Q_j \frac{P_j}{C_j} \left[\frac{P_i}{C_j} \right]^{\frac{1}{\rho}-1}. \quad (\text{A.6})$$

B Equilibrium Conditions

In the current appendix, we outline how we solve for the equilibrium.

Since the focus is on home firms' choice of intermediate goods, it suffices to concentrate on the home economy. We specify the supply of intermediate goods by foreign firms but suppress other details of the foreign economy, which are symmetric to those of the home economy. For households, the budget constraint is

$$\int_{j=0}^1 Q_j P_j = L_H. \quad (\text{B.1})$$

As discussed in the main text, by solving the utility maximization problem of households and the profit maximization problems of firms, we obtain the following equations: (2), (3), (4), (6),(7), (8), (17), (14).¹⁶ And the total expenditure of firm j on intermediate goods is

$$e_j = P_j Q_j = \int_{i=0}^N e_{ji} di. \quad (\text{B.2})$$

Similarly to the producer of intermediate good i in the home country, the maximization problem of foreign intermediate firms implies the pricing rule

$$P_{Fi} = \frac{1}{\rho \phi_{Fi}}. \quad (\text{B.3})$$

Using the entry and exit rates of intermediate goods firms in steady state, we obtain the measures of foreign and total intermediate goods in the home economy, which are shown in (33) and (35). Notice that the average profit of domestic intermediate firms in the foreign economy is

$$\int_{j=0}^1 \bar{\pi}_{Fji} dj = f_{fp} \times \left(\frac{\int_{j=0}^1 \bar{e}_{Fji} dj}{\int_{j=0}^1 e_{Fji} (\hat{\phi}_{Hif}^*) dj} - 1 \right), \quad (\text{B.4})$$

We can obtain the cutoff productivity level for the foreign economy $\hat{\phi}_{Hif}^*$, which satisfies

$$\pi_{Fi}(\hat{\phi}_{Hif}^*) = \left(\int_{j=0}^1 Q_j C_j^{\frac{\rho}{1-\rho}} dj \right) (\rho \hat{\phi}_{Hif}^*)^{\frac{\rho}{1-\rho}} (1 - \rho) - f_{fp} = 0. \quad (\text{B.5})$$

Combining all the equations listed in this Appendix B, we can calculate the equilibrium solution.

The equilibrium value of P_i is determined by equations (14), (17) and (B.3). And other endogenous variables ($P_j, Q_j, X_{ji}, P, Q, C_j, e_j, e_{ji}$) are determined by equations (2), (3), (4),

¹⁶To simplify the computation, we assume that the share of labor employed in the intermediate good sectors relative to total labor is identical domestically and internationally.

(6),(7), (8), (B.1), (B.2). The variables N_F and N are then determined by (30), equations (33), (35) and (B.4).

Based on the model, we link the change in welfare to the reduction in trade costs. Referring to equations (2) and (B.1), we observe that welfare depends solely on P_j . Furthermore, P_j equals C_j , which is determined by equation (7). From Proposition 1, we know that a decrease in domestic trade costs increases the number of varieties of intermediate goods N . As N increases, the price of the final good j decreases, as shown in equation (7). Therefore, total welfare increases as trade costs decrease.

C Measure of Intermediate Goods

In the Melitz model, cutoff productivity and average profit are core variables that affect the equilibrium number of firms. In Appendix C, we investigate how changes in domestic trade costs influence cutoff productivity and average profit, and thereby determine the number of intermediate goods firms.

When domestic trade costs decrease, the trade-cost-adjusted productivity governed by the Pareto distribution G in the equation (31) will rise, causing the FE curve to shift to the right. As shown in Figure E1, when the FE curve shifts from FE1 to FE2, the cutoff trade-cost-adjusted productivity increases and the average profit of domestic intermediate goods firms in the home economy $\bar{\pi}_{Hji}$ decreases. According to equation (35), this ultimately leads to an increase in the number of intermediate goods firms.

D Probabilities for Selection of Home and Foreign Producers of Intermediate Goods

Here, we present the calculation of the probability that the home producer of intermediate good i is favored over the foreign counterpart, $\mathbb{P}(\hat{\phi}_{Hi} > \hat{\phi}_{Fi})$, and the probability for the reverse situation, $\mathbb{P}(\hat{\phi}_{Fi} > \hat{\phi}_{Hi})$.

From the main text, we know $\hat{\phi}_{Hi}$ and $\hat{\phi}_{Fi}$ follow the Pareto distributions

$$G_H(\hat{\phi}_{Hi}) = 1 - \left(\frac{\hat{\phi}_{Hi,min}}{\hat{\phi}_{Hi}} \right)^{\alpha_H}, \quad \hat{\phi}_{Hi} \geq \hat{\phi}_{Hi,min}, \quad (\text{D.1})$$

$$G_F(\hat{\phi}_{Fi}) = 1 - \left(\frac{\hat{\phi}_{Fi,min}}{\hat{\phi}_{Fi}} \right)^{\alpha_F}, \quad \hat{\phi}_{Fi} \geq \hat{\phi}_{Fi,min}, \quad (\text{D.2})$$

where $\hat{\phi}_{Hi,min}$ and $\hat{\phi}_{Fi,min}$ are the lower bounds of the support. α_H and α_F are the shape parameters.

We assume the lower bounds of the support for home and foreign countries are the same; thus, the probabilities governing the selection between home and foreign producers of intermediate good i are given by the following expressions:

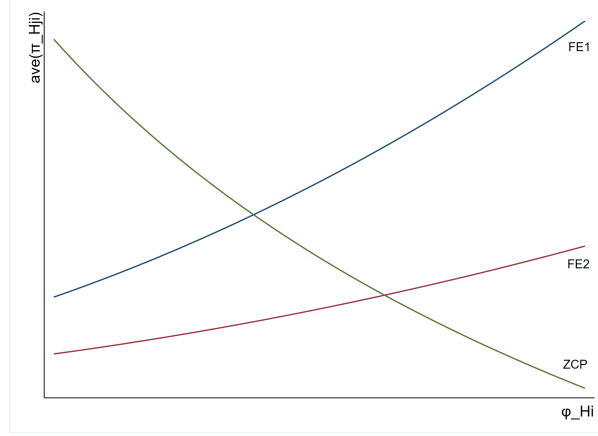
$$\mathbb{P}(\hat{\phi}_{Fi} < \hat{\phi}_{Hi}) = 1 - \frac{\alpha_H}{\alpha_H + \alpha_F}, \quad (\text{D.3})$$

$$\mathbb{P}(\hat{\phi}_{Hi} < \hat{\phi}_{Fi}) = \frac{\alpha_H}{\alpha_H + \alpha_F}. \quad (\text{D.4})$$

Given the parameter values of the Pareto distribution function, these probabilities are constants.

E Additional Figures

Figure E1: Domestic Trade Costs and Productivity and Profit of Intermediate Good Firms in Home Country



Note: This figure illustrates the change in equilibrium productivity and profit for intermediate good firms in the home country that is driven by the decrease in domestic trade costs. The y-axis represents the average profit of the home country's intermediate goods firms, and the x-axis represents productivity. ZCP refers to Zero Cutoff Profit. FE1 and FE2 denote the Free Entry Conditions in the original equilibrium and the new equilibrium, respectively. These two conditions (Zero Cutoff Profit and Free Entry Conditions) are calculated from the model. We set the Pareto shape parameter as 1.5, $\rho = 0.51$, $f_{hp} = 3$, $\delta f_e = 1.1$ and domestic trade costs decrease from 0.6 to 0.